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INTERNET PROTOCOL (IP) OVER LINK-16

THESIS

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AFIT/GCE/ENG/03-04

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AFIT/GCE/ENG/03-04

INTERNET PROTOCOL (IP) OVER LINK-16

THESIS

Presented to the Faculty

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Graduate School of Engineering and Management

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Air University

Air Education and Training Command

In Partial Fulfillment of the Requirements for the

Degree of Master of Science in Computer Engineering

Clinton W. Stinson, B.S.

Captain, USAF

March 2003

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AFIT/GCE/ENG/03-04

INTERNET PROTOCOL (IP) OVER LINK-16

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Clinton W. Stinson

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AFIT/GCE/ENG/03-04

Abstract

The purpose of Link-16 is to exchange real-time tactical data among units of the United States and allied forces. Primary Link-16 functions include exchange of friendly unit position and status data, the dissemination of tactical surveillance track data, and the control/management of air, surface, and subsurface engagements. Because Link-16 will play an integral part in the network-centric Joint Battlespace Infosphere (JBI), the performance of Internet Protocol version six (IPv6) and IP Security (IPSec) over Link-16 needs to be determined. IP packets also afford additional security measures within the JBI.

Using OPNET® modeling software to simulate a Link-16 network, the investigation of this research revealed that the overhead from IPv6 and IPSec does not significantly affect end-to-end delay and effective throughput of the Link-16 network. As long as the encryption and authentication protocols are preprocessed, these protocols add minimal amounts of latency overhead to the Link-16 network. However, as the offered load is extended beyond the 90 % level, the overhead from the IPSec extensions begins to have more of a negative effect on the End-to-End delay and throughput. Therefore, as the offered load increases beyond the 90 % level, it begins to have a significant impact on the performance of the Link-16 network.

INTERNET PROTOCOL (IP) OVER LINK-16

I. Introduction

One of the key challenges of the 21st century military force is Information Superiority. This challenge is being addressed in one respect through the Joint Battlespace Infosphere (JBI) [SAB99]. The JBI uses a network-centric concept, versus platform-centric concept, so that all JBI data can be easily transmitted from one platform to another. The JBI can accommodate both legacy and new communications systems. The integration of new and legacy systems provides essential improvements in the distribution of information through various platforms at all levels of the command structure from Joint Forces Air Component Command (JFACC) to the pilot in the cockpit [Ray01]. Link-16, a tactical data link used among U.S. and NATO forces, has the potential to bridge new and legacy systems through the use of the Internet Protocol (IP).

1.1 Background

The proposed JBI includes elements from deployed U.S., allied, and coalition forces that require the ability to communicate with one another [SAB99]. There is a relationship between the timeliness of information and the tempo of operations across any war-fighting theatre of operations. For instance, at the high end of the performance spectrum are cooperative sensing and engagement of high-speed targets that require high data rate and low latency information transport capabilities. At the intermediate level, there are various command and control activities that can tolerate information delays on the order of seconds. These operations are typically supported by

data links on various platforms such as fighter and support aircraft, fixed and mobile ground units, and naval vessels.

The JBI structure can be viewed as an integrated network of communication devices of multi-mode transport capabilities to include civilian and military networks, satellite communications, multiple types of data links, radios, and other commercial information services combined to create a distributed computing environment. Emerging technologies enable multiple stand-alone networks to be integrated into a dynamic network-of-networks communications system. In the current environment, voice, video, and data networks operate independently in order to meet required timelines for information exchange. Each network operates with protocols that are separate and distinct from the protocols employed in Transmission Control Protocol/Internet Protocol (TCP/IP) based networks, such as the Secret Internet Protocol Router Network (SIPRNET), or the Unclassified Internet Protocol Router Network (NIPRNET). Until recently, the reason for separate networks was due to lack of quality of service across IP networking technology. However, technology now exists to solve this problem.

Most current platforms use a tactical data link of one form or another. The Air Force is migrating its legacy data link systems to the J-Series family of tactical data links using Link-16 as the foundation. Link-16 will replace the Interim JTIDS Message System, TADIL-A, TADIL-B, and TADIL-C systems [USAF01]. The standard way of transporting data across most networks is through the use of IP packets. Therefore, it is essential that Link-16 be able to transport IP packets across its network as well. IP Next Generation (IPv6) is the latest version of the Internet protocol, designed to be the successor of IP version 4 (IPv4). Although one of the major reasons for creating IPv6 was to increase the IP address size from 32 bits to 128 bits, and thereby relieve the rapidly

shrinking available addresses, another key feature of IPv6 is its authentication and privacy capabilities. Extensions to the IP to support authentication, data integrity, and data confidentiality are specified in IPv6 [RFC 2460]. These extensions include the IP Security (IPSec) protocol which provides various security services at the IP layer. The two traffic security protocols contained in IPSec are the Authentication Header (AH) protocol and the Encapsulating Security Payload (ESP) protocol. IPSec is designed to provide interoperable, high quality, cryptographically-based security for IPv4 and IPv6 [RFC2401].

Although Link-16 has its own security measures (Message Security and Transmission Security), if additional security can be added without significantly adding to transmission overhead, then it is advantageous to provide security at an additional layer such as the IP layer. This research focuses on the latency effects from transmitting IP and IPSec over a Link-16 data link.

1.2 Goals

The overall goal of this research is to *evaluate the performance of a scheme that incorporates IP and IPSec into a Link-16 datalink network.* In assessing this goal, this research will first consider, as a baseline, the effect of IP overhead when "packaging" IP messages into JTIDS packets. Once the baseline is established, then the effects from the IPSec overhead will be considered. The IPSec protocols to be evaluated include the AH and ESP protocols. Their effect on network latency will be analyzed to determine if their additional overhead will adversely affect the Link-16 data link network. In order to attain the above stated goal, the following objectives will need to be met:

- Develop or obtain a Link-16 network simulation model
- Verify the simulation model
- Determine what impact IP messages passed over a Link-16 network have on overall latency

1.3 Document Overview

This chapter provides an introduction and some background to the network-centric concept of battlefield communications and focuses in on the data-link aspect, particularly, the Link-16 network aspect. It concludes with the goals of this research. Chapter II provides background information in the areas of the JBI, Information Assurance, IP, IPSec, and various communications platforms. Chapter III contains the methodology this research used to approach the problem. Chapter IV describes the verification and simulation process of the OPNET® Link-16 model, as well as the accumulation of data and analysis of the results acquired from the OPNET® Link-16 model. Chapter V describes research conclusions and areas that should be considered for future study.

II. Literature Review

2.1 Introduction

This chapter examines the increasing need for Information Assurance (IA) within older generation aircraft communications systems and the unique challenges these systems face when communicating with newer communications systems. Currently fielded IA methods for embedded information systems were designed on systems that were limited due to their proprietary interfaces to other systems. In contrast, network-centric warfare depends on a reliable flow of information among systems, which are designed around open architectures and commonly used standards and products. Additionally, older generation aircraft were not built to support the high data throughput rate common in many applications used today, nor can they support the graphical interfaces commonly used in many applications. Consider, for example the F-15E, a 70's era aircraft, which still plays a vital role in the U.S. Air Force. It is designed to support data transfer rates in the kilobit per second range not the megabit or even gigabit per second range that modern systems currently use. Modern systems are capable of high data transfer rates. These systems, such as the F-22 Raptor Stealth Fighter, also have IA integrated into them by design [Loc03]. It is a challenge to integrate IA into older generation aircraft such as the F-15E, not only because of limited bandwidth problems, but also because of the inherent difficulty in integrating new technology into older systems.

2.2 Scenario

Figure 2.1 shows a scenario in the proposed Joint Battlespace Infosphere (JBI) [SAB99]. The JBI is made up of a complex, heterogeneous system of systems with globally distributed fixed

and deployed assets consisting of various servers, databases, gateways, and proxies. Communications networks include WANs, LANs, terrestrial and space-based assets and also include such resources such as SIPRNET and NIPRNET. Figure 2.1 is split into two parts: fixed assets represent Continental United States (CONUS) resources and deployed assets represent outside (OCONUS) resources. In this scenario, an F-15E flight operating within the Joint Battlespace Infosphere (JBI) is enroute to its pre-planned target [Ray01][1].

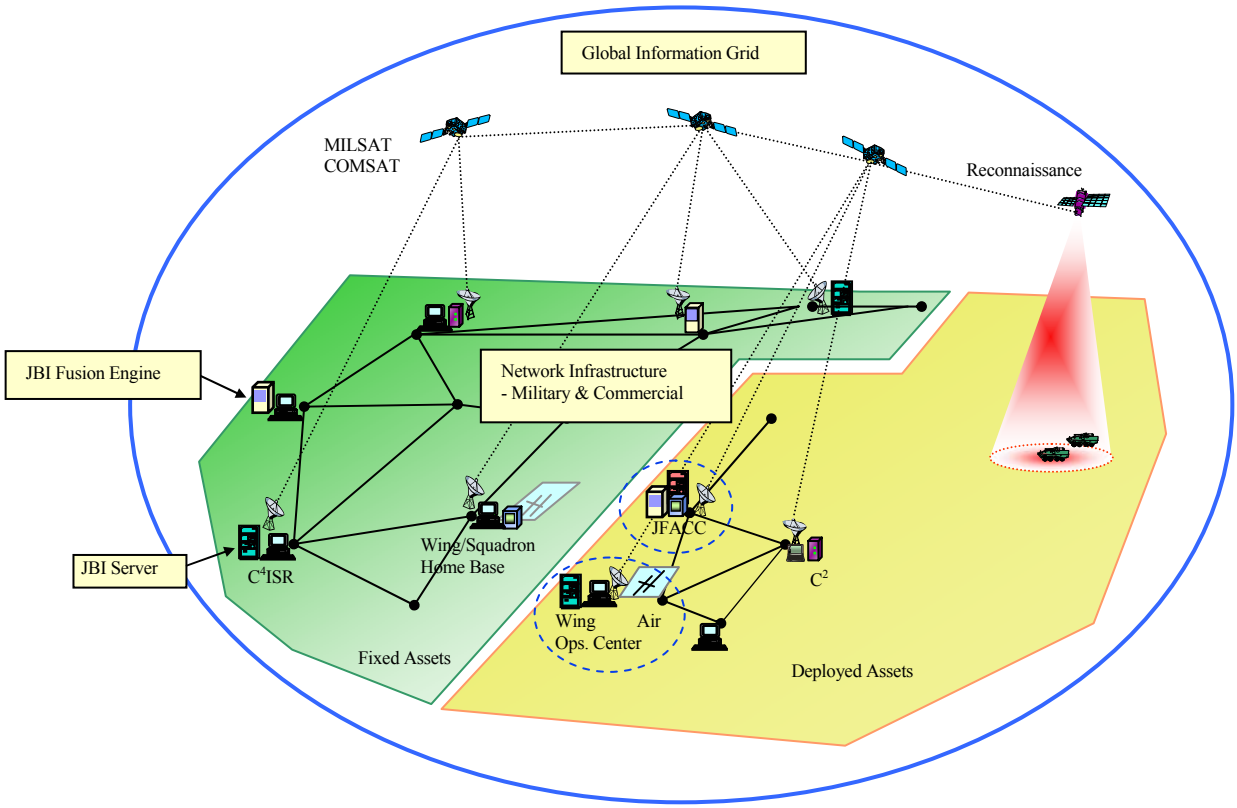


Figure 2.1: A Notional Deployed Joint Battlespace Infosphere

[1] Since the JBI is still a concept and not complete in design, assumptions have been made regarding ground-based JBI components.

Simultaneously, in the same theatre of operations, an Unmanned Air Vehicle (UAV) detects a Surface-to-Air Missile (SAM) and transmits this data to the JBI mission servers via Satellite Communication (SATCOM). Since SAMs are a high priority target, Air Command decides to re-route the F-15E flight to take out the SAM. Using JBI, Air Command directs the AWACS and F-15E to change the mission to intercept the SAM target. The F-15E on-board JBI client receives an Air Tasking Order (ATO) change alert. The AWACS operator and lead Weapons System Officer (WSO) review the ATO alert for additional info. The F-15E acknowledges the new ATO and diverts to the new target.

In this notional scenario there are many simultaneous communications occurring between fighter aircraft, AWACS, UAVs, satellites, JBI servers, and the Air Operations Centers (AOC), using various data formats, each encompassing their own security measures. It is problematic to insert data security into data communications due to the additional overhead that comes along with the added security. This is especially true for older generation aircraft with limited communications bandwidth. Yet, IA measures are needed to protect communication systems, data integrity, data confidentiality, data availability, and provide proper authentication and authorization measures.

Figure 2.2 shows an established F-15E—JBI communications link. Prior to the F-15E departure, the Link-16 network is configured to allow communication among the F-15Es, the AWACS controller aircraft and the ground-based AOC JBI Server Gateway. Once the connectivity between the on-board JBI client and ground-based JBI server is established, communication data is transferred via flight “Cups”, or objects whose implementation consists of a

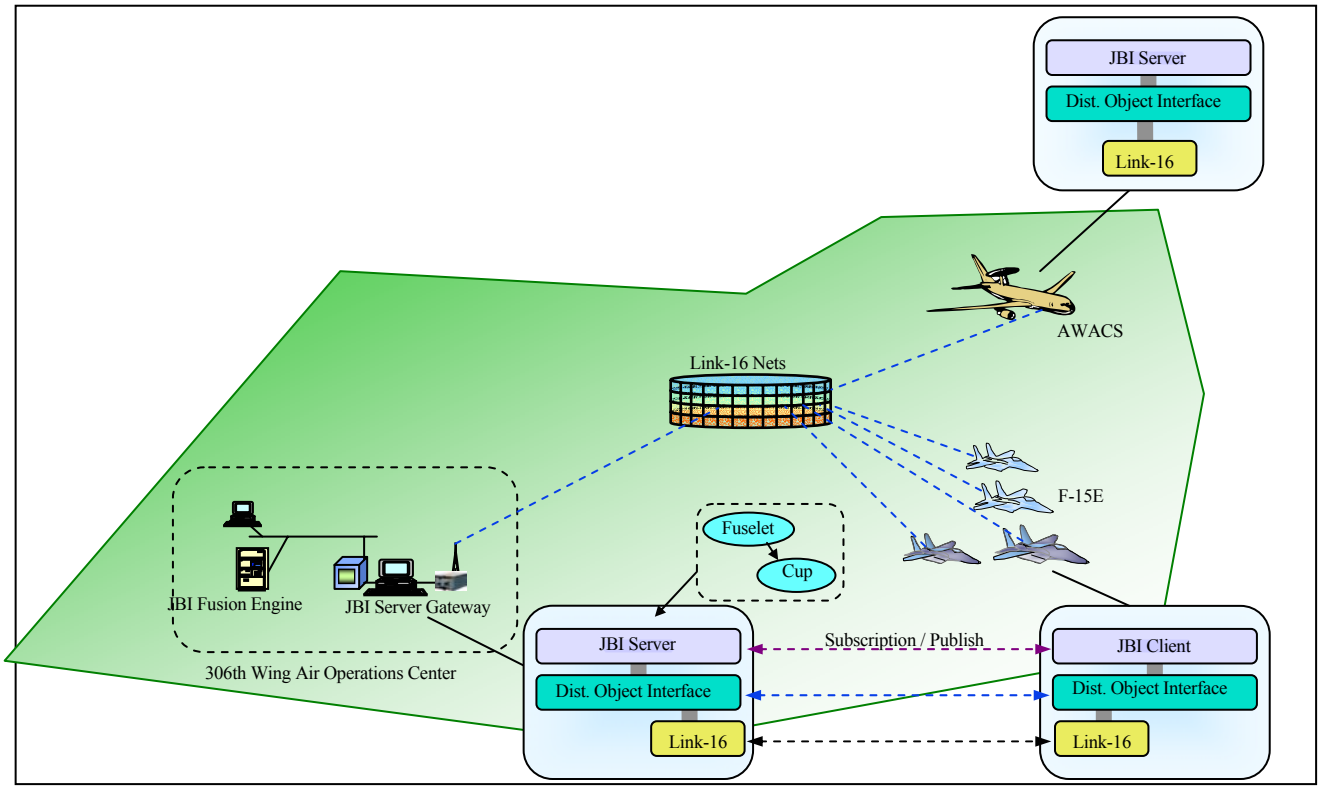


Figure 2.2: Linking the F-15E Aircraft into the JBI

CORBA object that provides services such as write and read to other objects. A CORBA object is defined as an identifiable, encapsulated entity that provides one or more services that can be requested by a client [TaV02]. Access to the Cup's services is typically restricted to objects that possess proper authorization rights.

Figure 2.3 shows the JBI Server Gateway that includes the JBI Server application and it's related databases and associated collaboration applications. The above mentioned "Cup" or mission fuselet resides on the JBI Server. In this scenario, CORBA serves as the distributed object

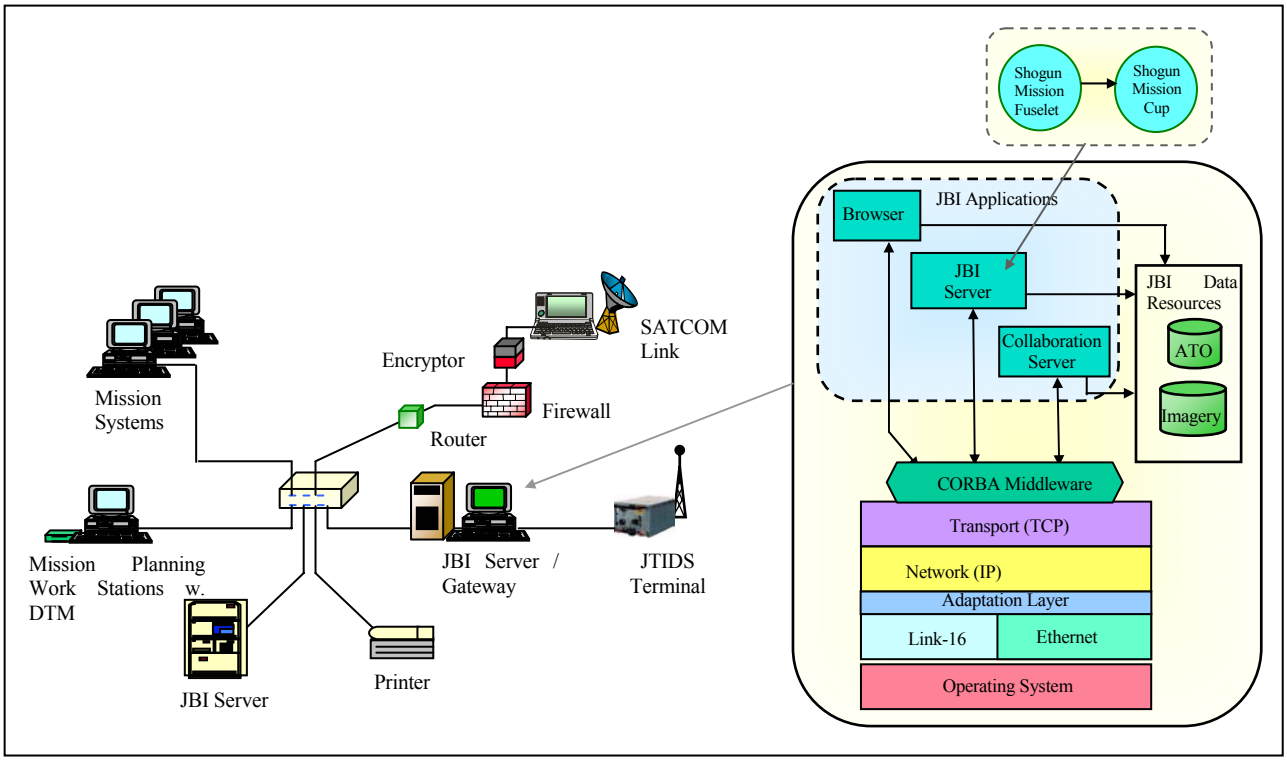


Figure 2.3: AOC Notional Hardware Architecture and JBI Server Gateway Software Architecture

middleware to facilitate communication between the different OSI layers in accordance with the security policy. An adaptation layer supports communication between the Internet Protocol (IP) and the Link-16 protocol. The F-15E on-board Advanced Display Core Processor (ADCP) serves as host to the JBI applications and required databases. Link-16 and SATCOM components communicate through the Communication, Navigation, and Identification (CNI) suite of the F-15E on-board communications system to provide wireless JBI connectivity. The CNI is connected to the ADCP via the MIL-SPEC 1553 Avionics Bus.

2.4 Information Assurance (IA) Capabilities.

IA functionality can be implemented in various components of the JBI—F-15E architecture. Countermeasures are layered to provide "defense in depth", where each layer provides it's own layer of security. Combined system security, then, is reinforced by each layer. For instance, security could be deployed in the ground-based host systems, in the JBI network nodes, or on-board the F-15E Strike Eagle. In focusing on the F-15E on-board notional architecture, there are several areas where IA functionality can be implemented, such as:

- Real-Time Operating System: Trusted Security Kernel, Access Control
- Data Link Layer: Link-16/JTIDS Security and SATCOM Security
- IP Layer: IPSec/IPv6.0
- Middleware Layer: CORBASec
- Application/Transport Layer: SSL and/or Database Security

This research focuses on security implemented at the IP layer. It is assumed that IPv6 will be used. IP Security (IPSec) is integrated into IPv6 and supports data origin, data integrity, data confidentiality, replay protection and automated management of cryptographic keys [Kae99].

2.4.1 Security Threats and Countermeasures. When considering implementation of IA into a communications system it is important to define threats and mechanisms available to counter those threats. There are four types of security threats to consider along with their typical countermeasures [TaV02]:

- Interception: This occurs when an unauthorized party gains access to data, such as when a third party eavesdrops on a conversation by two other parties or when data is illegally copied. A principle countermeasure to interception is data encryption.
- Interruption: Interruption occurs when data or services become unavailable, such as when data is corrupted or lost. A typical example is a Denial of Service (DOS) attack when a server can no longer be accessed because of overload. DOS is difficult to defend against, but authorization countermeasures put in place through a firewall are a typical method of protection
- Modification: Modification is the unauthorized changing of data or tampering with a service so it no longer conforms to the original specification. An example is tampering with database files or modifying the behavior of a program. Principle counter measures include authentication, authorization, and/or auditing. Authentication and authorization are put in place to prevent modification in the first place whereas auditing is used to identify a perpetrator after-the-fact.
- Fabrication: Adding information to gain unauthorized access, such as replay attacks or adding an entry into a password file are examples of fabrication. Similar to protecting against the modification threat, authentication, authorization and auditing provide a defense in this situation.

Some typical threats that might be encountered in the notional architecture include:

- Spoofing (modification): For example, communications from SATCOM to F-15Es can be modified, such that an attacker attempts to introduce data packets that appear to come from a trusted source. Countermeasures: TRANSEC and COMSEC of Airborne

Links and/or additional data encryption and packet source authentication mechanisms at higher communication layers.

- Introduction of malicious software into an F-15E from the JBI can occur when an adversary uses a JBI host to launch an attack against an F-15E. A typical situation is where a hacker finds a backdoor into the JBI network (through Battlefield networks, Defense Information System Networks, or the Internet) and creates a JBI object with embedded malicious code payload. The corrupted JBI object is uploaded to the F-15E aircraft via the Data Transfer Module (DTM), Link-16, or through SATCOM. The malicious code can then execute its payload on-board the F-15E. Countermeasures: Intrusion detection on-board the aircraft can mitigate the threat.
- Eavesdropping and/or Surveillance: This involves the unauthorized interception of information. Successful attacks against airborne links requires the ability to thwart TSEC and MSEC countermeasures at the data link level or the ability to monitor the unencrypted messages at the source or destination nodes. Countermeasures: Data encryption through TSEC and MSEC.

A comprehensive IA approach can be implemented by using a layered approach to security. These layers need to: (1) protect the system from attacks through access control, firewalls, and cryptography; (2) detect successful attacks (intrusions) through intrusion detection; (3) react to attacks by terminating the attack, confining and deleting malicious software, restoring the system to full integrity, notifying the pilot, WSO, and audit log.

2.5 Multi-Platform Common Data Link (MP-CDL)

MP-CDL provides a network-centric data link between airborne and surface Intelligence, Surveillance, and Reconnaissance (ISR) assets. The MP-CDL program (contract awarded in November 2002) is planned to meet the needs for a number of airborne and surface platforms to simultaneously distribute sensor data products to multiple supporting airborne and ground stations. MP-CDL is designed to meet the needs of various network clients (airborne and surface) to interact with a centrally located airborne terminal as well as other clients.

All terminals will support gateway connectivity to other links external to the MP-CDL network. These links may be either in-theater line-of-site (LOS) or beyond LOS such as SATCOM links. The initial application of MP-CDL will be in support of Army surface units command and control access to surveillance products from the Multi-Platform Radar Technology Insertion Program (MP-RTIP) platform. In addition to network operations, the MP-CDL terminals support the capability for point-to-point interoperability with CDL surface and/or airborne terminals.

The requirement for a central airborne terminal is to provide a single point-to-point data link operating simultaneously with an independent multi-user network. The terminal's point-to-point data link must be interoperable with exiting CDL surface communication equipment and Airborne Information Transmission (ABIT) relay terminals at established standard data rates up to 274 Megabits per second.

The multi-user network will connect up to 32 users on a COTS based network architecture. Range will be dependent on size, weight, and power requirements and mission geometries to be determined later, but is estimated to be approximately equal to the maximum LOS from an

altitude of 40,000 feet, or approximately 275-350 feet, depending on the height of the receiving antenna near the earth's surface. The MP-CDL system will operate in the Ku band and will support future capability to operate in one or more alternative RF bands (i.e., X, Ku, Ka) to allow multiple simultaneous links [PIX02].

The MP-CDL vision grew out of the MP-RTIP, which was originally a Joint STARS radar upgrade. MP-RTIP was restructured in 2000 to develop a common modular scaleable radar in three sizes:

- Large: Wide Area Surveillance (WAS)
- Medium: NATO
- Small: Global Hawk

2.5.1 MP-CDL Main Goals. The goals of MP-CDL are to provide:

- Transparent communication between deployed platforms [Cha02]
    - IP based, per Global Grid Standards
    - Low-latency, wideband path
    - Common carrier for all types of traffic in IP packets
    - Same HW/SW for air, ground, and sea
- A "LAN hub in the sky"
- Tradeoff Data rate vs. Antijam

- Common and COTS/GOTS Hardware

2.5.2 MP-CDL Data Rates. Throughput rates vary depending on the current configuration of one-to-one communication devices such as AWACS to UAV, AWACS to Common Ground Stations (CGS), UAV to CGSs. However, within the multi-user network, data transmission rate capabilities are based on these minimum required rates.

The multi-user network data transmission rate from the central airborne terminal (host) to the CGSs (clients) is:

- 45 Megabits per second unjammed to CGS
- 2.2 Megabits per second in jam resistant mode
- Similar rates from ISR hub for air-air net

The multi-user network data transmission rate capability from the CGSs (clients) to the central airborne terminal (host) is:

- CGSs: low send data, limited power and antenna size. Will dynamically share a low-rate up-link.
- A ground station with more bandwidth and a bigger dish could reach aircraft with 40-60 Megabits per second.
- ISR platforms should reach 20-40-60 Megabits per second air-air depending on geography and dish size.

2.5.3 Standardization Issues. The DoD tactical message standard is TADIL-J (Link-16). Non-tactical standard is IP. MP-CDL terminals will transmit and receive IP packets, and will not be involved in the content, format, or protocol of the data (unless the packet is addressed to that particular terminal). MP-CDL message sizes vary in length from 100 bits to 100 Mbits, and message types and sensor data from the Air Force, Army and Navy, such as TADIL-J, Moving Target Indicator (MTI), Synthetic Aperture Radar (SAR), Signals Intelligence (SIGINT), Air Tasking Order (ATO), and Global Grid (GG).

2.6 IPSec/IPv6

IPSec, short for IP Security, is a set of protocols developed by the Internet Engineering Task Force (IETF) to support secure exchange of packets at the IP layer [Kae99]. IPSec has been deployed widely to implement Virtual Private Networks (VPN). IPSec is supported in IP version 4 (IPv4) and is mandatory for the next generation of IP, version 6 (IPv6). IPSec supports two encryption modes: Transport and Tunnel. Transport mode encrypts only the data portion (*payload*) of each packet, but leaves the header untouched. The more secure Tunnel mode encrypts both the header and the payload.

A compliant IPSec implementation must support the required set of Security Association (SA) bundle types as outline in Section 4.5 of the Internet Engineering Task Force (IETF) Request For Comments (RFC) 2401 [RFC 2401]. The bundle types include four different combinations of the Authentication Header (AH) and Encapsulating Security Payload (ESP) protocols. On the receiving side, a compliant device decrypts each packet. The compliant protocol is designed to support these security areas:

- Data Origin Authentication
- Data Integrity
- Data Confidentiality
- Replay Protection

The Security Association (SA) concept is fundamental to IPSec. An SA is a relationship between two or more entities that describes how the entities will use security services to communicate effectively. The SA includes: an encryption algorithm; an authentication algorithm; and a shared session key.

For a compliant IPSec implementation to work, sending and receiving devices must share a public key. This is accomplished through a protocol known as *Internet Security Association and Key Management Protocol/Oakley (ISAKMP/Oakley),* which allows the receiver to obtain a public key and authenticate the sender using digital certificates [Kae99].

IPSec uses the AH protocol and ESP protocols to provide proof of data origin on received packets, data integrity, anti-replay protection, data confidentiality and limited traffic flow confidentiality. These two protocols can be combined and used to protect an entire IP datagram or just the upper-layer protocols of the IP payload.

Besides support for mobility, security is a key requirement for the successor to today's Internet Protocol version. Except for application-level protocols like SSL or SSH, all IP traffic between two nodes can be transmitted without changing any applications. All applications on a

machine that benefit from encryption and authentication policies can be set on a per-host (or even per-network) basis, not per application/service.

2.6.1 Authentication Header Protocol. Use of the AH protocol will increase the IP protocol processing costs and will also increase the communications latency. The increased latency is due to the additional authentication data contained in the AH. The fields of the AH as shown in Figure 2-4 are explained as follows:

- Next Header – 8 bit field which identifies the type of the next payload after the AH
- Payload Length – 8 bit field which specifies the length of the AH in 32 bit words
- Reserved – 16 bit field which must be set to zero
- Security Parameters Index (SPI) – A 32 bit value that in combination with the destination address identifies the SA for the datagram
- Sequence Number Field – Unsigned 32 bit field contains a monotonically increasing counter value for defense against replay attacks
- Authentication Data – Variable length field that contains the Integrity Check Value (ICV) for the payload

Next Header – 8 bits	Payload Length – 8 bits	Reserved – 16 bits
Security Parameters Index (SPI) – 32 bits		
Sequence Number Field – 32 bits		
Authentication Data – Variable Size		

Figure 2.4: Authentication Header (AH) Format

2.6.2 Encapsulating Security Payload Protocol. Use of the ESP protocol will also increase processing costs and communication costs in a similar manner as the AH protocol. The ESP header holds encryption, replay, and authentication information for its IP datagram. If authentication is selected as part of the SA, encryption is performed first followed by authentication. The encryption algorithm used is selected by the SA. ESP is designed to use symmetric key encryption algorithms. The fields of the ESP header as shown in Figure 2-5 are:

- Security Parameters Index (SPI) – A 32 bit value that in combination with the destination address identifies the Security Association for the datagram
- Sequence Number Field – Unsigned 32 bit field contains a monotonically increasing counter value for defense against replay attacks
- Payload Data – Variable length field containing data described by the Next header field. If the encryption algorithm requires an initialization Vector then that would be contained here
- Padding (0-255 bytes) – May be required to satisfy requirements for encryption algorithms
- Pad Length – Indicates the number of bytes used in the Padding field
- Next Header – Identifies the type of data contained in the Payload field
- Authentication Data – Variable length field that contains the Integrity Check Value (ICV) for the packet

<table>
<tr><td colspan="3">Security Parameters Index – 32 bits</td></tr>
<tr><td colspan="3">Sequence Number – 32 bits</td></tr>
<tr><td colspan="3">Payload Data – Variable Size</td></tr>
<tr><td colspan="3">Padding – 0 to 255 bytes</td></tr>
<tr><td></td><td>Pad Length</td><td>Next Header</td></tr>
<tr><td colspan="3">Authentication Data – Variable Size</td></tr>
</table>

Figure 2.5: Encapsulating Security Payload (ESP) Format

2-7. Link-16.

The purpose of Link-16 is to provide a mechanism for the exchange of real-time tactical data among units of U.S. forces, Joint Forces, and NATO forces. Link-16 is an integral part of the Joint Battlespace Infosphere (JBI) and provides the network backbone for JBI communications. Link-16 provides Message Security (MSEC) through Type 1 Encryption, and provides Transmission Security (TSEC) through frequency hopping.

Link-16 uses the Joint Tactical Information Distribution System (JTIDS) for the communications component of Link-16. The JTIDS data terminal encompasses the Class-2 terminal software, hardware, RF equipment, and the high-capacity, secure, anti-jam waveform that they generate [Nor01]. The JTIDS terminal is an advanced radio system that provides for the rapid exchange of tactical information among a large number of users. The U.S. Air Force Class-2 terminal implements the Class 1 Interim JTIDS Message Specification (IJMS) protocol as well as JTIDS. JTIDS operates in the L_x band between 960 MHz and 1215 MHz and employs the Time

Division Multiple Access (TDMA) architecture. By using different frequencies, a technique called "frequency hopping", multiple nets can be "stacked" through the simultaneous use of time slots. Each time slot is 7.8125 milliseconds in duration. JTIDS provides 51 different frequencies for frequency hopping. The frequencies assigned to JTIDS for TDMA[1] transmissions vary in range from 969 MHz to 1206 MHz in 3 MHz increments. Each pulse is transmitted on a different frequency in a pseudorandom pattern that depends on the net number and the TSEC crypto-variable. The nominal frequency-hopping rate is greater than 33,000 hops per second [Nor01].

2.7.1 Data Exchange Rates. Link-16 can transmit either 3, 6, or 12 data words in a 7.8125 msec (1/128 sec) time slot depending on whether the Standard, Packed-2, or Packed-4 data packing structure is used. Each Link-16 data word is made up of 75 bits, of which 70 bits are data, 4 are used for parity and 1 is reserved as a spare bit. The *effective* tactical data rates of Link-16 are 26.88 kilo bits per second (kbps), 53.76 kbps, or 107.52 kbps, depending on the data packing structure used. Each 7.1825 msec time slot of unencoded information holds 450 bits at Standard packing, 900 bits at Packed-2 packing, and 1800 bits at Packed-4 packing. Because error detection and correction (EDAC) requires 16 bits for every 15 bits of data, the same time slot with Reed-Solomon encoding only holds 210, 420, and 840 bits of tactical information for the Standard, Packed-2, and Packed-4 encoding [Nor01].

[1] Link-16 operates on the principle of Time Division Multiple Access (TDMA), wherein 128 time slots per second are allocated among all participating JTIDS Units (JU) for the origination and reception of data.

Therefore, the effective tactical data rates are calculated by multiplying the number of tactical information bits per slot by the number of slots per second, for example:

- Standard packing: 210 data bits (3 × 70 bits/word) × 128 = 26.88 kbps
- Standard packing with 5 parity bits: 225 (3 × 75 bits/word) × 128 = 28.80 kbps
- Standard packing with EDAC: 465 (3 × 155 bits/word) × 128 = 59.52 kbps

Therefore, when using EDAC, a little less than half of the JTIDS word is used for data. If EDAC is not used, then the entire word can be used for data.

If the additional bits for Reed-Solomon EDAC encoding are considered, then data rates[1] increase to 59.52 kbps, 119.04 kbps, and 238.08 kbps [Nor01]. Link-16 supports Link-4A and Link-11[2] functions as well as additional functions such as voice, navigation, and an expanded electronic warfare capability. The table below shows a comparison to Link-11 and Link-4A common data rates.

Table 2.1: Link-16 Data Rate Comparison

				Data Rates (kbps)		
Link	**Architecture**	**Protocol**	**Message Standard**	**Tactical**	**With Parity**	**With EDAC**
Link-4A	TDM	Command/ Response	V-Series R-Series	3.06	--	--
Link-11	Netted	Polling by Net Control	M-Series	Fast – 1.80 Slow – 1.09	--	2.250 1.365
Link-16	TDMA	Assigned Time Slots	J-Series	Standard – 26.88 Packed-2 – 53.76 Packed-4 – 107.52	28.8 57.6 115.2	59.52 119.04 238.08

[1] These data rates should not be considered "effective" data rates because they include encoding overhead, however they are provided to give the reader an idea of actual "non-effective" bandwidth that Link-16 is capable of achieving.

[2] Link-16 provides communication improvements over its ancestors, the Link-4A (TADIL C) and Link-11 (TADIL A/B) tactical data link architectures.

2.7.2 Link-16 Data Security. Link-16 encrypts both the message and the transmission. Message security (MSEC) uses the KGV-8 encryption device and cryptovariables to encrypt message traffic. Transmission security (TSEC) is also accomplished through the use of cryptovariables, which control the JTIDS waveform. An important feature of the waveform is its use of frequency hopping. The hopping pattern is determined by both the net number and the TSEC cryptovariable. The TSEC cryptovariable also determines the amount of jitter in the signal, and a predetermined, pseudorandom pattern of noise that is mixed with the signal prior to transmission.

2-8. Common Object Request Broker Architecture (CORBA)

CORBA, considered to be the most widely-used middleware standard, is an industry-defined specification for distributed systems. The CORBA specifications are a product of the Object Management Group (OMG) [TaV02]. The global architecture (reference Figure 2.6) of CORBA consists of four groups of architectural elements connected to what is call the Object Request Broker (ORB).

The ORB forms the core of any CORBA distributed system and is responsible for enabling communication between objects and their clients [Tav02]. In the notional architecture discussed above, CORBA objects (Cups) may reside either in the JBI server or on-board the aircraft. Security threats to CORBA objects may appear in the form of attackers attempting unauthorized access to a CORBA object or unauthorized creation of a CORBA object. Countermeasures include the use of ORBs, which fully support the CORBA Security (CORBASec) specification, at the CORBA middleware level of the notional architecture as shown in Figure 2.6.

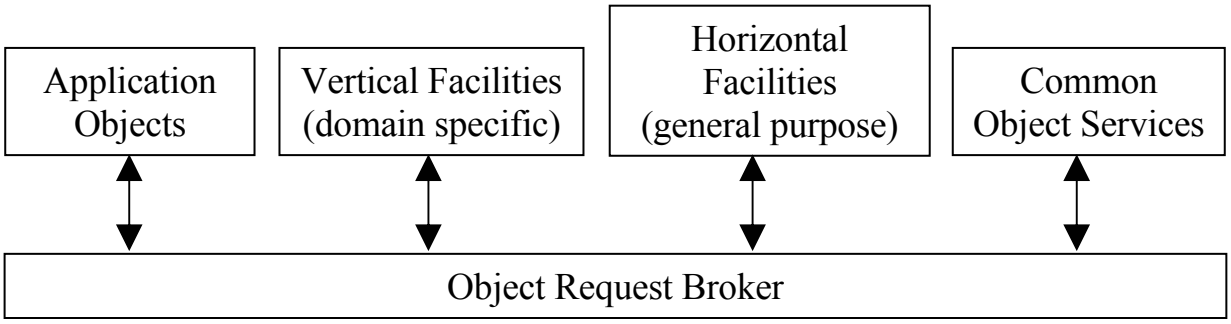


Figure 2.6: The Global Architecture of CORBA

2-9. Current Research.

The following summarizes other research efforts either ongoing or completed related to this work.

2.9.1 Tactical Digital Information Link (TADIL) J Range Extension (JRE). Link-16 has proven to be an effective mode of communication for Tactical Data Links and will continue to be an integral part of military weapon's communications systems. The major draw back to Link-16 is it is limited to LOS communications. However, there are several initiatives working to extend JTIDS beyond LOS. The most common method of extending JTIDS range beyond LOS is through the employment of airborne relays. However, this is not always possible due to lack of airborne assets. Other means of extending beyond LOS include use of satellite communications, or use of a program called Joint Range Extension (JRE) [DGSP97].

The Joint Data Network (JDN) capacity is often strained because many participants require relays to reach Non LOS units. A solution has been proposed using SATCOM as a means to supplement TADIL J traffic. The primary means of distributing data in the JDN is with the JTIDS system. JTIDS uses a TDMA architecture for the transmission and reception of voice and data on

a finite number of time slots. When relays are needed to reach NLOS units, the number of time slots is doubled for that particular operation.

The TADIL J JRE program proposes a possible solution to the relay problem. The program is being conducted in three phases. The Phase 1 demonstration is a simple check to see if TADIL-J messages could be sent through a satellite and received within the required TMD latency. Phase 1 was successful and demonstrated that TADIL-J messages could be relayed through a satellite in near real time. Phase 2 of JRE was similar to Phase 1 but the data was passed through a STU-III before it was reformatted into a J3.6 message before being sent via SATCOM. Phase 2 also proved successful. Phases 1 and 2 were conducted without using JTIDS terminals or networks. The Phase 3 demonstration connected remote JTIDS networks through the satellite range extension. The Phase 3 demonstration was also successful and demonstrated there was a potential savings of time slots on the JDN utilizing the JRE. JRE also provides more reliable connectivity in hostile environments because airborne assets might not be available.

2.9.2 ATM Network-Based Integrated Battlespace Simulation With Multiple UAV-AWACS-Fighter Platforms. This research provides a realistic input to the amount of throughput that is required to support realistic C4I applications, real time battle management, SAR image processing and analysis, and real time air tasking order (ATO) monitoring through the demonstration of an integrated battlespace simulation on an advanced AWACS prototype network. The integrated battlespace simulation includes the unmanned aerial vehicle (UAV), C4I platform, and fighter aircraft as core battlefield components. The simulation uses a scenario not unlike the scenario introduced at the beginning of this chapter. For its demonstration it uses both ATM LAN emulation and classical IP over ATM multicast configurations.

ATM Classical IP-based (CIP) Multicast Solution: The ATM CIP protocol lacks a broadcast mechanism. This is resolved by setting up point-to-multipoint permanent virtual circuit (PVC) connections from a broadcast server to all clients. Since there is no such server in CIP, creation of a virtual broadcasting node (that corresponds to a broadcast service access point) at the switch is necessary.

ATM LAN Emulation-based Multicast Solution: ATM LAN emulation can support multiple independent emulated LANs (ELANs), and the membership in any of the ELANs is independent of the physical location of the end system. The AWACS mission computer must be a member of all ELANs so that it can selectively broadcast information to any of the AWACS, fighter, or UAV group as different multicast groups.

The maximum throughput in the TCP stream test was 108 Mbps for the ATM LAN emulation solution and 118 Mbps for the ATM CIP solution. As long as the socket buffer sizes were kept above 64 Kilobytes, then both solutions demonstrated normal operation.

2.9.3 IP Mobility Management for the Airborne Communications Node (ACN) Platform. In network-centric architecture where data is transferred via IP packets, it is important to consider issues that occur when moving from one ACN to another. These include: average signal strength; subscriber mobility as they move from one footprint of an ACN to another; and other types of mobile subscribers. A potential problem occurs when, for example, an entire brigade moves relative to the ACN, into the footprint of another ACN. Mobility management must ensure that routing to and from the edge routers continues to operate correctly. One solution is Mobile IP, an IETF protocol designed to handle IP mobility [RFC 2002]. However, the range of movements of

the edge routers will be restricted to within the ACN network domain. Therefore, only small-scale mobility management is required and an alternative to the Mobile IP solution can be considered such as a dynamic routing table update solution. Link state routing performed better than the mobile IP solution in terms of overhead and does not have a single point of failure (as in Mobile IP with its home agent) [JaW00]. Although security overhead was not considered, other overhead issues were which may be beneficial when considering security overhead [JaW00].

2.9.4 Surveillance and Control Data Link Network (SCDLN) for Joint STARS. The Joint Surveillance and Target Attack Radar System (Joint STARS) communications systems is used to connect an airborne radar platform and many mobile Ground Station Modules (GSM) [SaB94]. The SCDLN uses a secure, highly jam-resistant, dynamically alterable two-way digital data link for the control and distribution of information. The SCDLN has an additional capability to provide an autonomous message communication network over a wide aerial coverage in a hostile environment. The major contributor to the anti-jam performance is the Fast Frequency Hopping (FFH) spread spectrum waveform. Of particular interest in this article is the network architecture. Messages are transmitted in packets and the format will support the transfer of TADIL-J (JTIDS) packet messages.

The network is configured so that an airborne platform retransmits incoming data from any GSM back to another or multiple GSMs within the local theatre of operations. The network operates in half-duplex mode where time is divided into bursts, each burst lasting for 100 milliseconds. The downlink operation (from airborne platform to GSM) occupies approximately half this time, and two independent uplinks plus a guard time occupy the other half. The retransmittal of the uplink message becomes an automatic acknowledgement to the sending GSM

that the uplink message was correctly received at the AWACS and also allows addressing information to any other GSM in the network (or all of them). An unlimited number of GSMs can copy both downlink sensor data and relayed messages. However, a maximum of 15 GSMs can be active at any time and participate in transmitting uplink messages. Any GSM or AWACS can be the master in the network at a given time. GSMs are allowed to enter or depart from the network. Once the AWACS commences downlink transmission, an initial polling sequence begins and the network is established by each GSM searching independently for the AWACS downlink signal and once found, begins downlink tracking.

2-10. Summary.

This chapter provides a general background and literature review for this research. A notional JBI architecture was presented and how Link-16 fits into the overall JBI scenario was discussed. Of particular interest are IA issues within the JBI and the best approach to establishing a robust IA security within the Link-16 arena through the use of IPv6 and IPSec. Other approaches were considered through the use of CORBA security using objects to control access rights.

III. Methodology

3.1 Problem Definition.

As communications technologies have developed, military systems have migrated from stand-alone systems to client-server and fully networked systems. Therefore, more stringent security requirements have resulted in increased demands on security mechanisms. The integration of embedded systems such as the F-15E within the proposed Joint Battlespace Infosphere (JBI) network topology exposes this aircraft to new information warfare threats. Exacerbating the problem, information assurance technologies designed for use in real-time embedded systems have not kept pace with emerging threats [Ray01]. Link-16 is a prime candidate to provide a network backbone for the proposed JBI communications. The Joint Tactical Information Distribution System (JTIDS) terminal, which makes up the communications component of Link-16, provides Message Security (MSEC) through data encryption and Transmission Security (TSEC) through frequency hopping. However, further security can be implemented through the network layer of the communications architecture.

3.1.1 Goals and Hypothesis. The research goal is to evaluate the performance of an Information Assurance scheme that incorporates IPv6 and IPSec over a Link-16 datalink network. This goal is further defined by the following sub-goals:

1. Evaluate the performance metrics of a baseline Link-16 system that incorporates IP packets across the Link-16 network.
2. Determine the impact of incorporating IPSec into the baseline system.
3. Determine the impact of various offered loads to the baseline system.

It is hypothesized that IPv6 and/or IPSec can be incorporated into the Link-16 network without degrading performance to a level that it is incapable of supporting real-time data and voice transmission.

3.1.2 Approach. To accomplish the above stated goals, a Link-16 model was used with the OPNET® network simulation software to simulate IP traffic over a Link-16 network. A distributed software system was used in which an external model communicates with the OPNET® software to simulate incoming JBI traffic fed to the Link-16 network. This system provided the necessary model to compare IP baseline traffic to IPSec to determine effects of increased load on the Link-16 network.

3.2 System Boundaries.

The System Under Test (SUT), Figure 3.1, includes the F-15E JBI Connectivity Software Architecture. This includes the Real-Time OS, the Physical Layer (Link-16), an Adaptation Layer, the Network (IP) Layer, the Transport (TCP) Layer, Real-Time CORBA Middleware, and the JBI applications. Also included but not pictured in Figure 3.1 is the JTIDS terminal, the communications component used to transmit data and voice transmissions across the Link-16 network. Not included in the SUT are the other components of the F-15E Avionics System Architecture which include Intelligence, Sensors, and Radar (ISR) collecting components. Within the SUT, the Component Under Test (CUT) includes the JTIDS terminal, the Physical Layer (Link-16), the Adaptation Layer, the Network Layer (IP), and the Transport Layer (TCP).

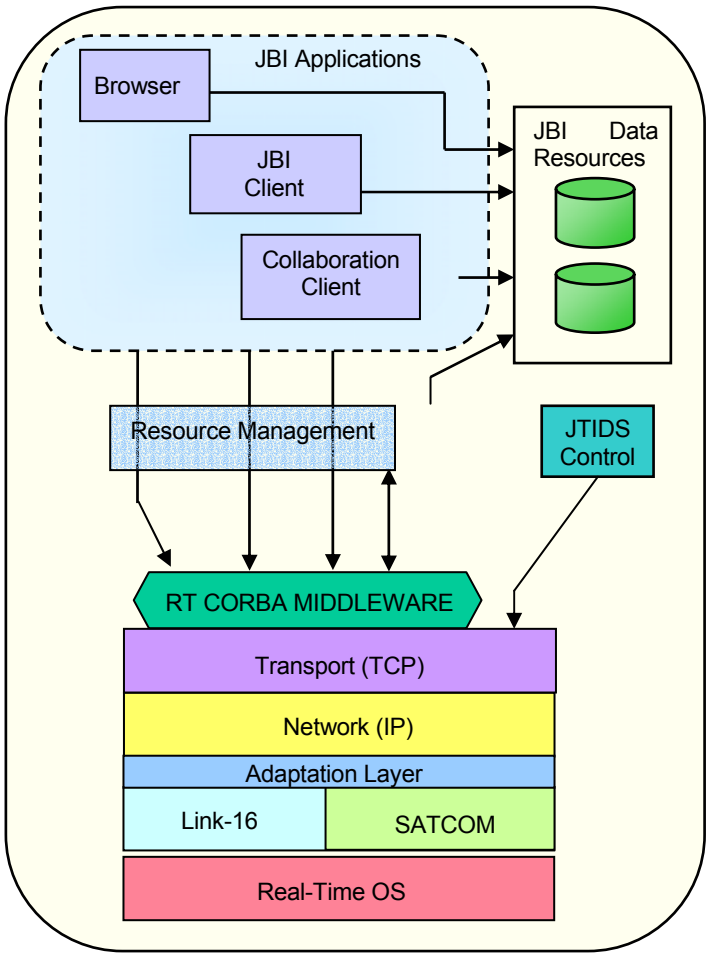


Figure 3.1: F-15E JBI Connectivity Software Architecture [Ray01]

3.3 System Services.

The network layer decouples upper layers with independence from the data transmission and switching technologies used to connect systems. In addition, the network layer provides network security including security services at the IP layer of the TCP/IP protocol stack. The set of security services IPSec can provide includes data origin authentication, data integrity, confidentiality (encryption), and rejection of replayed packets (a form of partial sequence integrity), and limited traffic flow confidentiality. All of these services are provided at the IP

layer and can be used by any higher layer protocol such as TCP and UDP [Kae99]. The primary services for this study and their possible outcomes include:

1. Data Origin Authentication.
    a. Success – Packet received from valid origin (packet accepted)
    b. Failure – Unable to establish that packet came from valid origin (packet dropped)
2. Data Integrity, Data Confidentiality.
    a. Success – Packet payload has not been tampered with (packet accepted)
    b. Failure – Unable to establish that packet payload can be trusted (packet dropped)
3. Replay Protection.
    a. Success – Packets are prevented from being resent from an unauthorized source therefore preventing unauthorized access (packet dropped)
    b. Failure – Unable to detect that a packet came from an unauthorized source (packet accepted)

3.4 Performance Metrics.

The following metrics are used:

1. Throughput – Throughput is defined as Transfer Size/Transfer Time, where Transfer Size is measured in bits and Transfer Time is measured in seconds. Transfer Size is defined to be number of *tactical* data bits, thus does not include Reed-Solomon encoding. The effective tactical data rates of Link-16 vary depending on the data packing structure used.

2. End-to-End (ETE) delay – ETE is measured in seconds and is defined to be the elapsed time from when a packet arrives at the source node's routing layer to when the packet is received by the routing layer of the destination node.
3. Data transfer – In Link-16, either 3, 6, or 12 Link-16 words can be transmitted in a 7.8125 msec time slot depending on whether the Standard, Packed-2, or Packed-4 data packing structure is used. For instance if the Packed-2 packing structure is used, then 420 (six 70-bit words) data bits are transferred in 1/128 second to give an instantaneous rate of 52.5 kbps.

3.5 Parameters.

Within the system boundaries, the system and workload parameters that affect performance are defined below.

3.5.1 System Parameters.

1. JTIDS Transmission/Reception Equipment – This equipment is the communications component of Link-16. It encompasses the Class 2 terminal software, hardware, RF equipment, and the high-capacity, secure anti-jam waveform that they generate [Nor01].
2. JBI Applications – These include the JBI server, browser, and collaboration server.
3. JBI Data Resources – These include Air Tasking Orders, Imagery, etc.
4. Network Layers – TCP, IP, physical layer.
5. RT CORBA Middleware – Intermediary for passing objects between TCP layer and JBI applications.

3.5.2 Workload Parameters.

1. Data rate – The tactical data rate range for Link-16 is 28.80 kbps to 115.20 kbps.
2. Voice and data workload – A trace of traffic measured on a real-time system will be used to compare against simulated results. Data includes tactical data information such as navigation waypoints, target assignment, target tracking, release points, munitions inventory, and sensor data.
3. IPv6 versus IPSec – IPv6 provides the channel for transferring IP packets. IPSec provides the additional security measures required for multi-level security within the F-15E notional architecture.
4. Operating System (OS) – Proprietary real-time OS versus COTS.
5. Packing structure – Either Standard, Packed-2, or Packed-4 data packing structure is used.

3.6 Factors.

The following factors and their corresponding levels were chosen as the most significant for this research.

3.6.1 Data rate.

1. Standard Tactical Rate (with parity) – 28.8 kbps
2. Packed-2 Tactical Rate (with parity) – 57.6 kbps
3. Packed-4 Tactical Rate (with parity) – 115.2 kbps

3.6.2 Internet Protocol.

1. IPv6 – This provides a baseline performance analysis of IP packets over Link-16.

2. IPSec – Additional overhead from IPSec is considered to determine if the Link-16 network can handle the increased workload. The two protocols of concern are the Authentication Header protocol and the Encapsulated Security Payload protocol

3.7 Evaluation Technique.

An OPNET® Link-16 model developed by the Navy Space and Naval Warfare (SPAWAR) Systems Command office is used to evaluate this performance analysis. Some modifications to the model were made by the Air Force Research Lab (AFRL) Sensors Directorate. An external mission model was used with the OPNET® Link-16 model to input mission data such as message type, message ID, message size, source, and destination.

3.8 Workload.

Workload parameters are based on the Link-16 OPNET® model as listed in Table 3.1:

Table 3.1: Workload Parameters

Workload Parameter	Setting
Data Rate	57.6 kbps
Offered Load	30, 60, 90 %
Message Length	450 bits
Type of Message	J-series type
Packing Structure	Packed-2 (six words per time slot)
Security Level	AH, ESP, or both
Network Participation Group (NPG)	Varies according to mission function

3.9 Experimental Design.

This experimental design consists of specifying the number of experiments, the factor level combinations for each experiment, and the number of replications of each experiment. Since this experiment included one factor with three levels and another factor with three levels, there were $3 \times 3 = 9$ experiments. Five replications were conducted for a total of $5 \times 9 = 45$ simulations. The 30%, 60%, and 90% offered load levels were chosen to show the effects these loads have on the Link-16 network when its lightly loaded, moderately loaded, and heavily loaded. Although three different security levels were chosen (AH, ESP, and a combination of AH and ESP), it was not necessary to repeat a full set of simulations for the ESP level since it only adds two more bytes of overhead to the IP packet and the difference in the simulation results was negligible. Therefore, a few simulations were run to verify that the difference was negligible when adding 26 bytes of overhead from the AH extension versus adding 28 bytes of overhead from the ESP extension.

3.10 Summary.

This chapter described the methodology to be used for the performance analysis of implementing IP packets over the Link-16 network. It discussed the thesis goal, approach to be used, system boundaries and services, performance metrics, parameters, factors, and workload.

IV. IMPLEMENTATION AND ANALYSIS

4.1 Overview

This chapter provides research results and analysis. The verification and validation of the OPNET® implementation of the Link-16 model is described and a description of the baseline IPv6 over Link-16 model is presented. For comparison to the baseline, security features are added to IPv6 to further test the Link-16 network's capacity to handle increased demand. Finally, this chapter provides an overview of results and overall analysis.

4.2 Link-16 Verification and Validation

OPNET® is a Commercial Off-The-Shelf (COTS) program that provides an environment for network simulations. It is widely used throughout the DoD for network modeling and provides support for detailed radio modeling, which is a key requirement for JTIDS. OPNET®'s network traffic is generated stochastically, using probability density functions. Therefore, generated data packets do not contain information, but are just tokens that represent data of a given size that transverse a given network. This is sufficient for this study, since we are only interested in how overhead and data load affect overall performance, the particular information contained in packets is irrelevant. The model need is based on a model provided by the AFRL Sensors Directorate and uses OPNET®'s radio propagation model, referred to as the Radio Transceiver Pipeline, and simulates the protocol message packet and models a JTIDS terminal's transmissions on a time slot basis. Figure 4.1 shows the communications system consisting of an external mission model, and the JTIDS hosts and JTIDS terminals used in the Link-16 model. The mission model is used to communicate with the Link-16 model and provides the offered load and mission data to the Link-16 network.

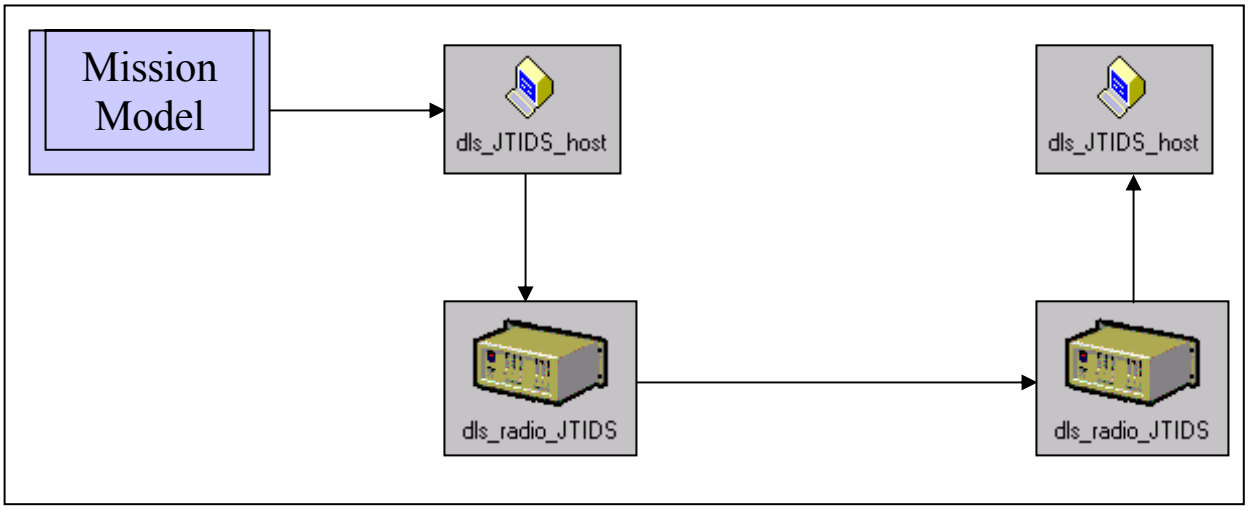


Figure 4.1: Mission Model – Link-16 Communication System

This OPNET® model was configured using the Link-16, Packed-2 packet format, which contains two 3-word blocks of 225 bits each for a total of 450 bits. For example, one particular test sent 2160 bytes (17,280 bits) from a Narrow Area Search Munitions (NASM) terminal to the Airborne Command and Control Center (ABCCC) terminal. The 17,280 bits were divided up and inserted into 39 (17,280/450 = 38.4) JTIDS packets.

The JTIDS transmitter terminal adds 35 bits of overhead to the 6-word JTIDS packet for a total of 485 bits. Average data rate is calculated based on 450 bits transmitted every 1/128th of a second, therefore 450 × 128 = 57,600 bps. The terminal model supports transmission and reception of free-text format messages, the format used to transmit IP packet data through the Link-16 network. The terminal model is a simplified representation of a JTIDS terminal that enqueues incoming TADIL-J messages from the host into available buffers, and sends them out in the correct timeslot. The terminal model uses the OPNET® Radio Transceiver Pipeline to calculate the effects of Radio Frequency (RF) propagation. The pipeline stages used are modified versions of the default radio pipeline stages provided by OPNET®. The modifications allow for

improved bit error rate calculation and add support for animation of the radio links. The host-to-terminal interface is represented as a duplex point-to-point link with zero delay. Although it is not representative of the 1553 bus, the latency is factored into the Link-16 model.

JTIDS scenarios require the terminal nodes and host computer pairs to be co-located in subnets for proper spatial movement. This organization is imposed by OPNET® because node models connected by physical links (as the terminal and host are by the dls_serial link) cannot be mobile.

Figure 4.2 shows the OPNET® host node model used to generate J-series messages, which are sent to the JTIDS terminal shown in Figure 4.3.

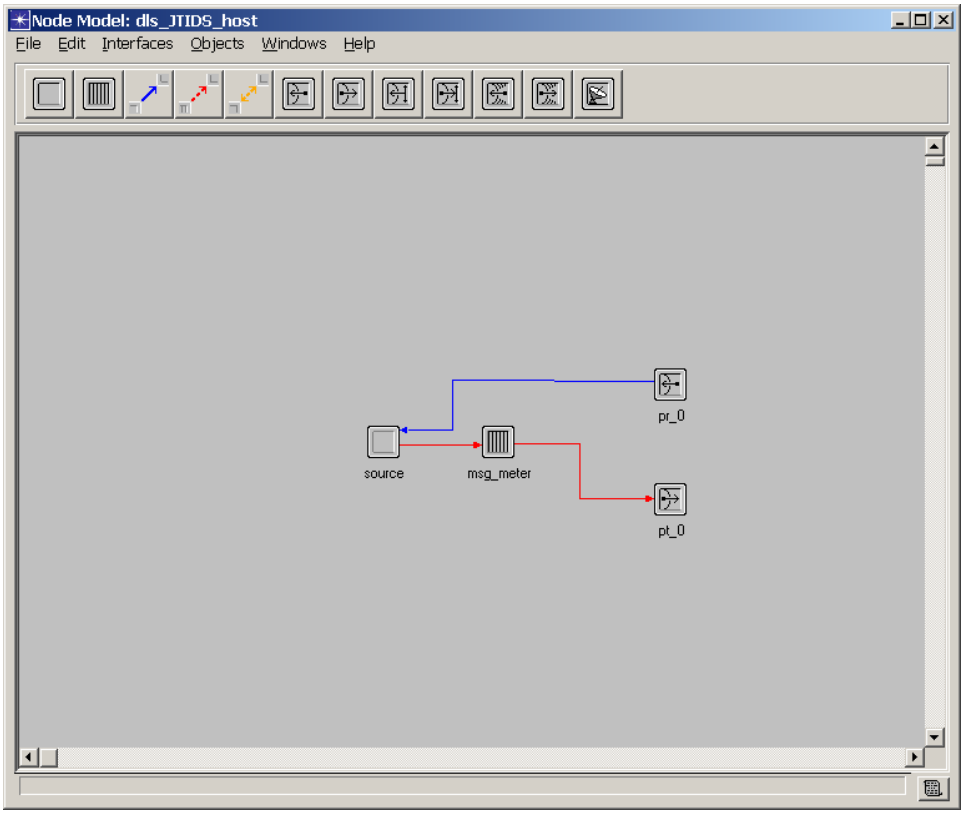


Figure 4.2: dls_JTIDS_host Node Model

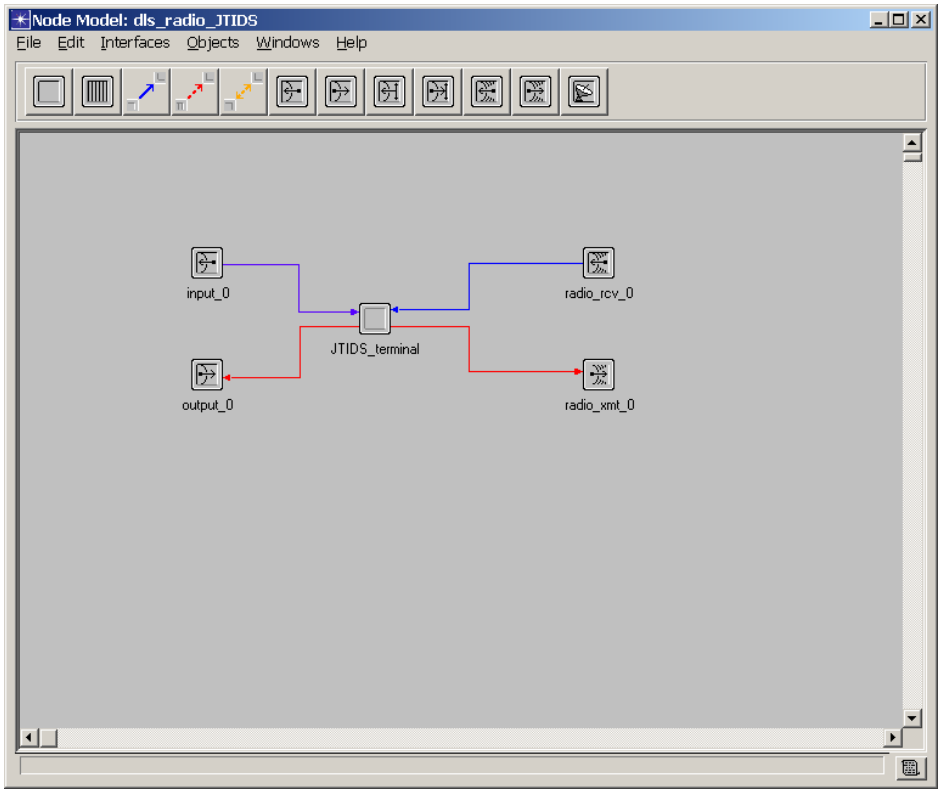


Figure 4.3: dls_radio_JTIDS Node Model

This J-series message packet format represents a single TADIL-J message consisting of six 75-bit words. The JTIDS terminal accepts these packets through the point-to-point serial interface connecting it to the host computer. The JTIDS terminal encapsulates one or more TADIL-J messages that are sent in the time slot, along with the message headers. The number of messages encapsulated is dependent on the packing type and the length of each TADIL-J message, which in the case of this model uses the Packed-2 format with a message length of 450 bits.

Validation was not accomplished on this model because no real-time data could be obtained to validate the results against.

4.2.1 Verification Implementation. The basic implementation of the JTIDS model used for verification included the parameter settings in Table 4.1. Performance metrics include the End-to-End (ETE) delay and throughput.

Table 4.1: Verification Workload Parameters

Workload Parameter	Setting
Data Rate	57.6 kbps
Offered Load	30 % (2,160 bytes or 17,280 bits)
Message Length	450 bits
Type of Message	J-series type
Packing Structure	Packed-2 (six words per time slot)
Security Level	None
Network Participation Group (NPG)	NPG-29

4.2.2 Sample size for determining mean. To estimate the system's mean performance with an accuracy of $\pm r$ % at a confidence level of $100 \times (1-\alpha)$%, the number of observations n required to achieve this goal is can be determined from Eq. (4-1) [Jai91]:

$$\text{Confidence Interval} = \bar{x} \pm z\frac{s}{\sqrt{n}} \tag{4-1}$$

Where z is the normal variate of the desired confidence level. The desired accuracy of r % indicates that the confidence interval should be ($\bar{x}$(1-r/100), $\bar{x}$(1+r/100)). Using this and Eq. (4-1) and solving for n yields Eq. (4-2):

$$n = \left(\frac{100zs}{r\bar{x}} \right)^2 \tag{4-2}$$

With an accuracy of 5 % and based on preliminary tests, each set of simulations requires $n \leq 5$ runs. Refer to Appendix D for calculations used to determine the required number of simulations.

4.2.3 Verification Results. The Link-16 OPNET® model was verified by presenting data at the host terminal and ensuring the data went across the host terminal to the sending JTIDS transmitter terminal, then to the receiving JTIDS terminal, and finally to the receiving host terminal. During this process, the number of bytes received at the receiving JTIDS terminal was compared against the number of bytes sent at the sending JTIDS terminal to ensure that they were equal.

The ETE delay was measured and compared against the ideal ETE delay. For instance, when sending 17,280 bits through the network with the data rate set at 57,600 bits per second, the ETE delay should be 17,280 bits/57,600 bps = 0.300. To verify the ETE delay, a MATLAB® program (cfi, Appendix E) was used to create exponentially distributed IP messages at a frequency of 17,280 bps (2,160 bytes per second) for the baseline model representing a 30 % offered load.

The MATLAB® software program was used to create a script file that was input into an external "mission model". The mission model and the OPNET® Link-16 model make up a distributed system in which the mission model uses the script file to input mission data to the OPNET® Link-16 model. It includes the following types of data: type of message sent, time message was transmitted, message ID, source ID, destination ID, and message quality. Originally, messages were sent based on a deterministic distribution, or in other words the

messages were sent every second with exactly one second between transmittals. However, this method didn't provide a realistic model of how traffic arrives at a network node and multiple runs generated the same results each time. Therefore, an exponential distribution was created in order to model the bursty nature of network traffic arriving at a node.

Through several simulations it was determined the OPNET® model reached steady-state after 3000 samples were run, as shown in Figure 4.4. The average ETE delay was just over 0.250 seconds. Although the average ETE delay is typically expected to be equal to, or greater than, the ideal ETE delay, the reason it is less can be explained by either one of, or a combination of the following:

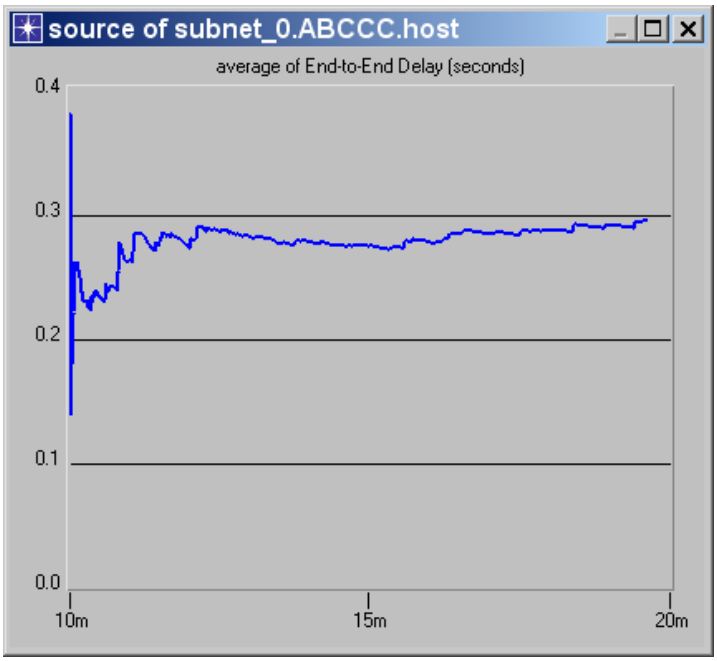


Figure 4.4: Average ETE Delay (2160 Byte Packet, 3000 Samples)

- The Radio Frequency (RF) data rate of the Link-16 model was set at 97.0 kbps, instead of 57.6 kbps to better approximate an effective throughput of 57.6 kbps, because the real terminal sends at a much higher rate to leave time within the time slot for propagation guard and jitter. The total time required to transmit the header and data portion of the timeslot in the Packed-2 Double-Pulse structure is 5.772 ms. Because the model is generic in the sense that it can support, Standard Double Pulse, Packed-2 Single Pulse, Packed-2 Double Pulse, or Packed-4 Double Pulse message packing, it uses an average time to model jitter and propagation. Therefore, depending on which type of data pulse is used, there will be some built in error.
- Although the exponential distribution was set up to send an IP packet to the host terminal every second, the actual calculated time between packets sent varies from 0.97 seconds to 1.03 seconds, causing the ETE delay and throughput to be offset by various amounts.

The OPNET® model was also checked to ensure it was using the proper packing configuration (Packed-2) and if it was adding the additional 35 bits of overhead to each 450-bit word.

Messages were traced and verified that a total of 485 bits were being transmitted through the JTIDS terminals. Through several simulations and debugging methods, the above mentioned verification tests confirmed that the OPNET® model was operating correctly.

4.3 JTIDS Baseline.

This research is studying the effects of IPv6 packets over the JTIDS network as well as the effect IPSec, with its additional security overhead bits, has on the latency and throughput of the

Link-16 network. The IPSec areas of interest are Authentication Header (AH) and Encapsulating Security Payload (ESP) protocols and are explained in the next section.

4.3.1 Baseline Implementation. The implementation of the Link-16 model used as a baseline includes the parameter settings outlined in Table 4.2. An external model was used to communicate with the OPNET® model in order to simulate IP traffic downloaded to a JTIDS host terminal.

Table 4.2: Implementation Workload Parameters

Workload Parameter	Setting
Offered Load	30, 60, or 90 %
IP Message Length	2160, 4320, 6480 bytes plus 28 bytes for AH and 54 bytes for AH and ESP
JTIDS Message Length	450 bits
Type of Message	J-series type
Packing Structure	Packed-2 (six words per time slot)
Security Level	AH, ESP, or Both
Network Participation Group (NPG)	NPG-29

4.3.2 Baseline Results (ETE Delay). The ETE is measured in seconds and is defined to be the elapsed time from when a packet arrives at the source node's routing layer to when the packet is received by the routing layer of the destination node. Figure 4.5 shows the ETE mean delay for offered loads of 30, 60, and 90 %. For the 30 % load, 2,160 bytes (17,280 bits) were

generated every second for an average ETE delay of 0.256 seconds. The ideal ETE mean delay should be 17,280 bits/57,600 bps = 0.300 seconds.

The 60 % offered load (4320 bytes) resulted in an average ETE of 0.652 seconds. The 90 % workload (6,480 bytes) resulted in an average ETE of 1.354 seconds. Additional simulations were run using 8,000 byte IP packets at which point the ETE average was 2.506. As expected, an increase in the number of bytes per second beyond the 90 % offered load caused the buffer to backup, and required several hours for the OPNET® simulations to reach completion.

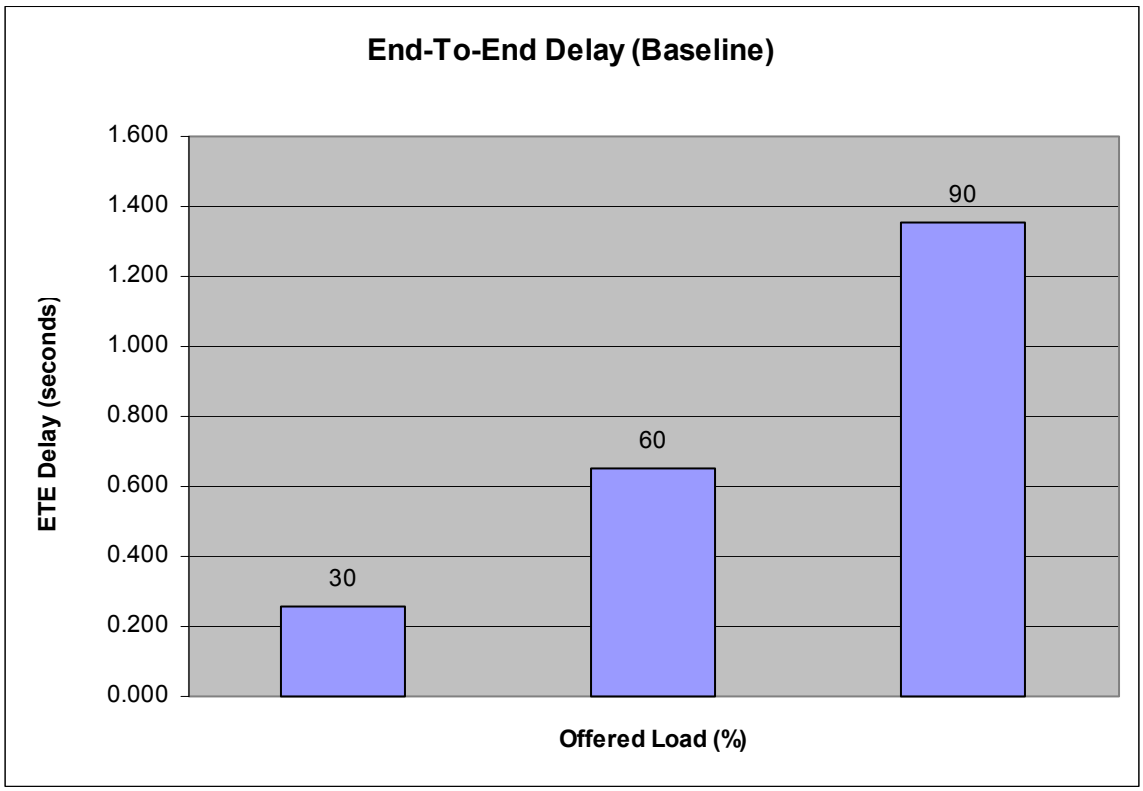


Figure 4.5: Baseline End-To-End Delay

4.3.3 Baseline Results (Effective Throughput). Figure 4.6 shows the effective throughput for nominally offered loads of 30, 60, and 90 %. Effective throughput was calculated by using the total number of bits offered at the host terminal divided by the total simulation time. The total

number of bits only includes the data bits, and does not include any overhead bits. The actual offered load varied slightly from the nominally offered load. For instance, the 30 % offered load should present 17,280 bits per second (bps) (57,600 × 0.30) to the host terminal. However, as mentioned above, the exponential distribution is used to calculate the time each IP packet is presented to the host terminal, but the actual calculated value for how often packets are presented may differ from the nominal valued by as much as several milliseconds. As was the case in the 30 % offered load, the actual distribution time was 17,280 bits per 1.02 seconds. Therefore the actual offered load was 16,941 (17,280/1.02). The measured effective throughput of 16,949 bps is slightly different due to rounding errors in the Link-16 model.

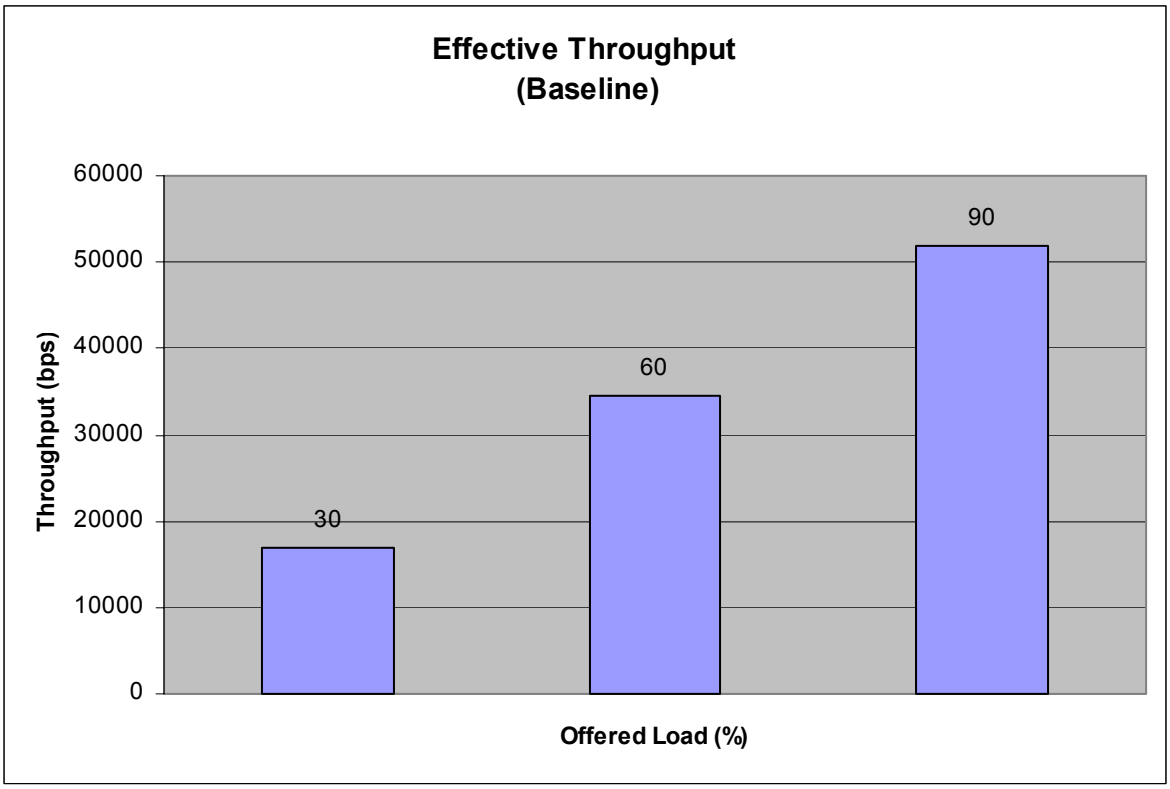


Figure 4.6: Effective Throughput – Baseline

At the 60 % nominally offered load level the Link-16 network is only receiving 34,491 bps and is not fully utilized and there is less chance of messages backing up in the buffer. However, at the 90 % workload level, the host terminal is receiving 51,798 bps, which is near its maximum capacity and messages back up in the buffer more frequently, thereby negatively impacting the effective throughput and thus explains the super-linear growth.

4.4 JTIDS Security Feature Additions (IPSec)

Security and privacy in IPSec is integrated into IPv6 and provided through the AH and ESP protocols.

4.4.1 Authentication Header (AH) Protocol – End-To-End Delay. The addition of the AH to the IPv6 baseline involves using the keyed MD-5 hash function, or any one way hashing algorithm to compute a 128-bit digest of the message to be transmitted [CDK01]. In this effort, the MD-5 has function is considered for calculating additional overhead bits. The 128 bits returned from the hash function is appended to the authentication data contained in the AH. As mentioned in Chapter Two, the AH adds on 12 bytes from the first five fields of the AH and another 16 bytes (128 bits) of authentication data for a total of 28 additional bytes added to the IP packet. Therefore, even though this process requires extensive computer preprocessing, it only adds 28 bytes of overhead to the baseline IP packet, thus effecting network latency and ETE delay very minimally as shown in Figure 4.7 [Kae99].

4.4.2 Authentication Header (AH) Protocol -- Effective Throughput. Figure 4.8 shows the effective throughput for offered loads of 30, 60, and 90 %. Similar to the ETE delay, the effective throughput is minimally impacted due to the addition of the AH protocol.

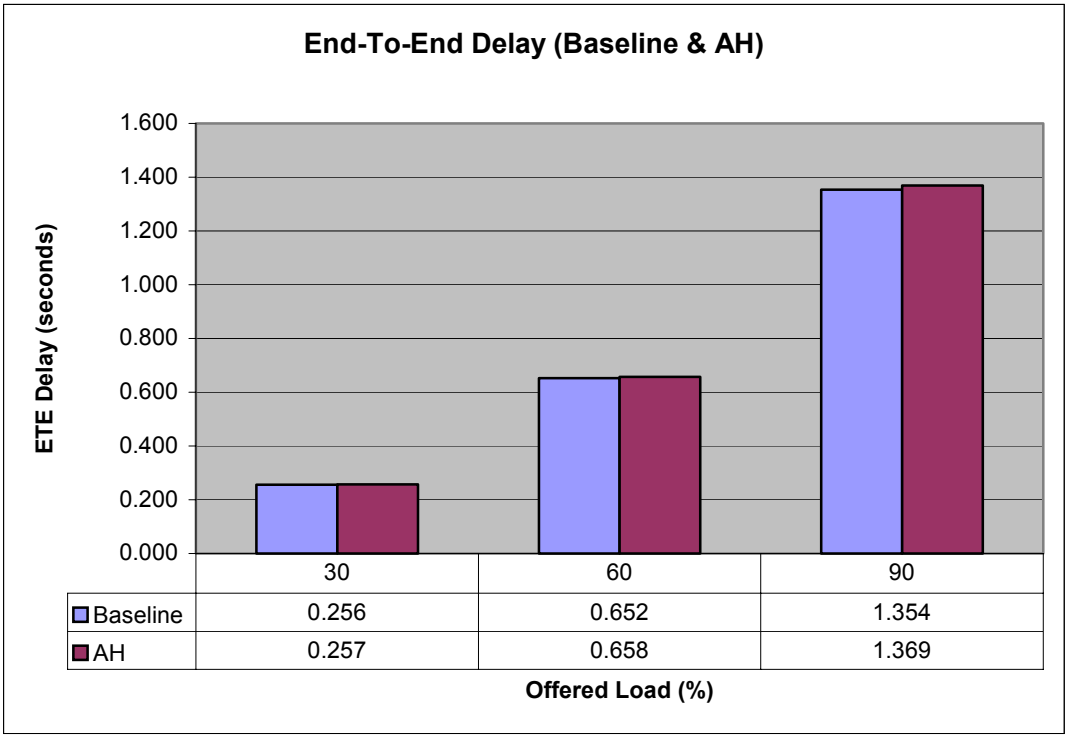


Figure 4.7: Baseline and Authentication Header -- End-To-End Delay

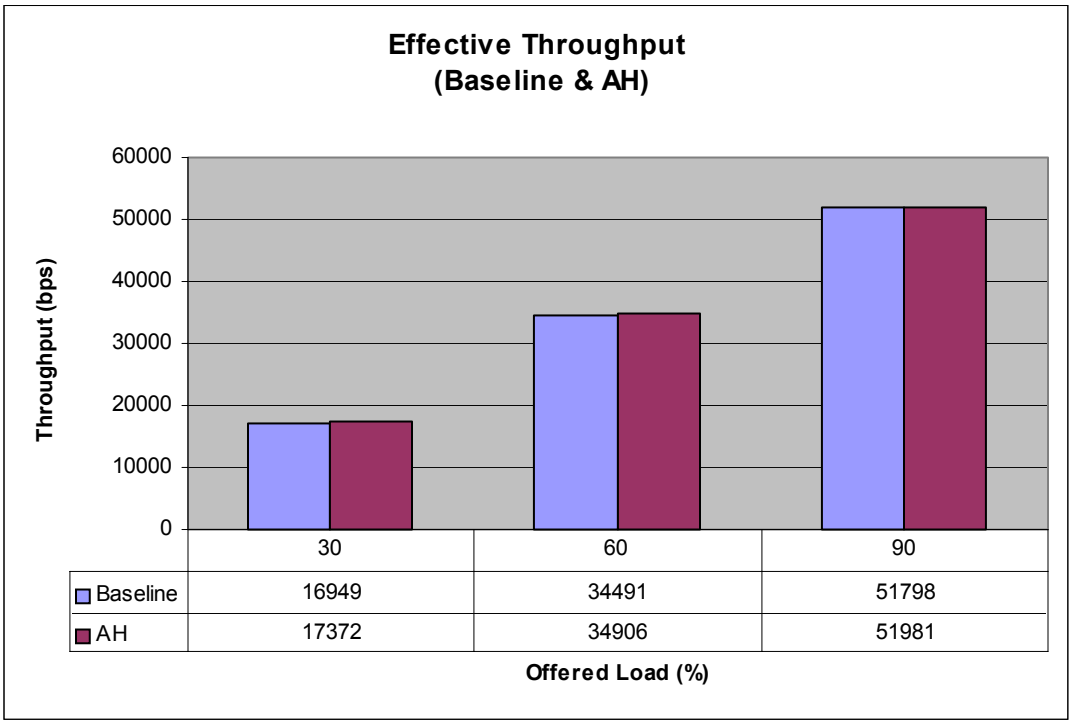


Figure 4.8: Baseline and Authentication Header -- Effective Throughput

4.4.3 Encapsulating Security Payload (ESP) Protocol – Effective Throughput. The addition of the ESP header to the IPv6 baseline uses an encryption algorithm to secure the payload. A common encryption algorithm used is triple DES. Triple DES uses a 112-bit key to encrypt the payload data, therefore adding 14 bytes onto the payload data. As described in Chapter Two, the ESP protocol adds a minimum of 12 bytes and up to an additional 255 bytes (if required by the encryption algorithm). A total of 26 bytes will be used as additional overhead data for the ESP protocol.

Since the additional 28 bytes of overhead from the AH had such a small effect on the overall ETE delay, it is reasonable that simulations ran using 26 bytes of overhead would be statistically equivalent and therefore, in the economy of time, a full set of simulations was not conducted for the ESP extension, but instead a few simulations were run to verify that results were not significantly different from the simulations run for the AH extension. The overall effect on throughput was also minimal.

4.4.4 Baseline, Authentication Header, and Encapsulating Security Payload -- ETE Delay. ESP and AH headers can be combined in a variety of modes. For these simulations, an average for the combination of the ESP and AH headers was used which added an additional 54 (28 + 26) bytes of overhead. As can be seen in Figure 4.9, the increase in overhead has a minimal effect on the average ETE delay. The 30 % offered load increases from 0.256 for the baseline, to 0.470 for the 60 % offered load, and 1.060 for the 90 % offered load.

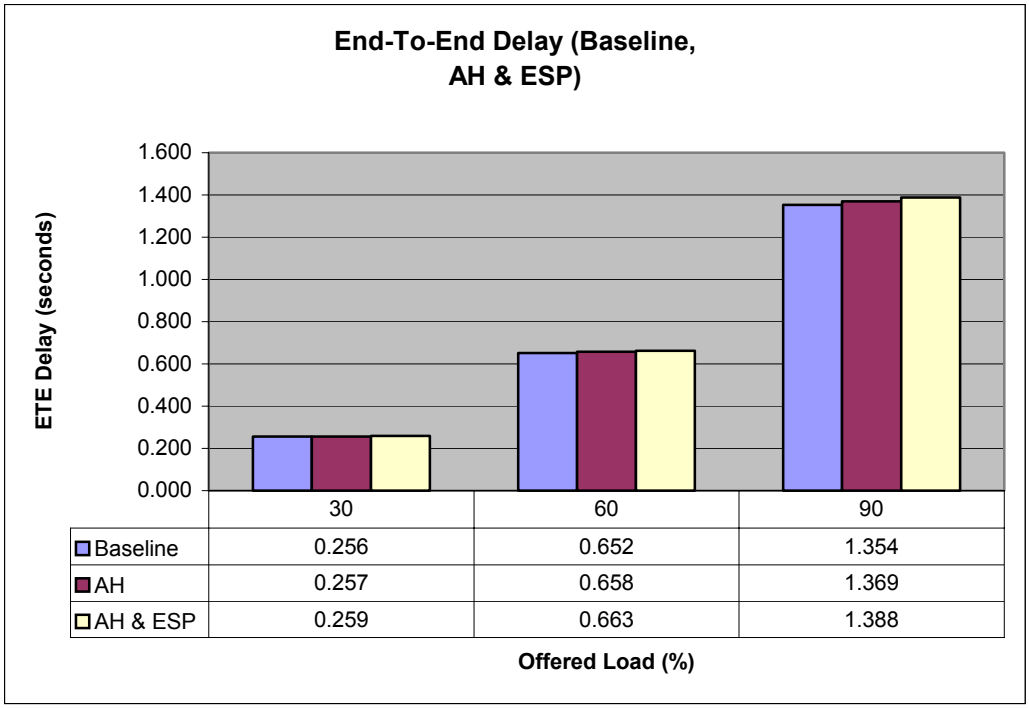


Figure 4.9: Baseline, Authentication, and Encapsulating Security Payload End-to-End Delay

4.4.5 Baseline, Authentication Header, and Encapsulating Security PayloadProtocols – Effective Throughput. Figure 4.10 shows the effective throughput for offered loads of 30, 60, and 90 %. The throughput for the 30 % workload increases slightly from 16,949 bps for the baseline, to 17,372 bps for AH, and again increases slightly to 17,582 bps for the combination of AH and ESP. The slight increase from the AH configuration to the AH and ESP combined configuration shows that the difference is statistically insignificant as can be seen in Table 4.3 because the F-Computed values are less then the F-Table values. The 60 and 90 % workloads follow a similar trend.

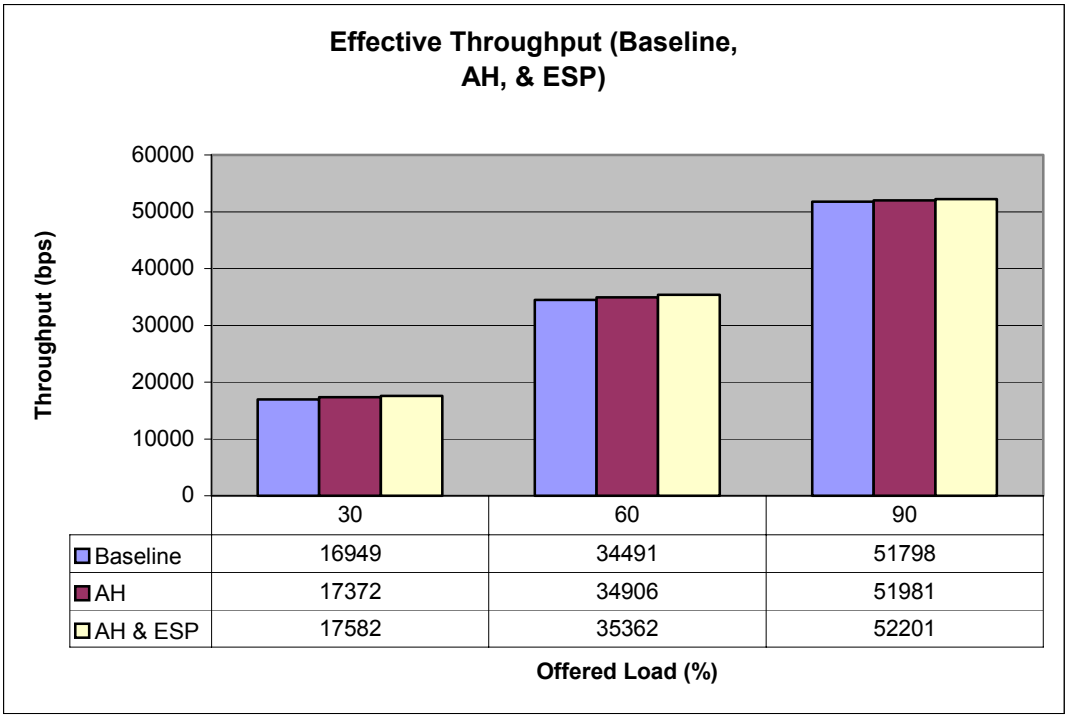


Figure 4.10: Baseline, Authentication, and Encapsulating Security Payload Effective Throughput

4.5 Result Analysis

The next sections present an analysis comparing the three overhead variations (baseline, AH, AH and ESP combined) to the offered load variations (30, 60, 90 % offered load). All the ANOVA tables and derivation formulas are also displayed together in Appendices A through C.

4.5.1 End-To-End (ETE)Delay Analysis. As seen in Figure 4-9 above, the ETE delay increases as packet size (offered load) increases. The variation shown in Table 4.3 shows that 99.94 % of the variation was due to the change in workload. Only 0.02 % of the variation was due to increased overhead. The variation due to overhead-workload interaction was 0.01 % and variation due to error was 0.02 %. Because the F-Computed value for offered load is significantly larger than the F-Table value, this confirms that offered load has a significant impact in Link-16 network performance. The F-Computed value for the overhead variation is slightly larger than the

F-Table value; therefore, it has a minimal effect on the network performance. Because the F-Computed value for interaction is less than the F-Table value, the overhead-offered load interaction does not have an effect on the network performance.

Table 4.3: End-to-End (ETE) Delay Allocation of Variation (ANOVA)

Table A.6: End-To-End Delay Allocation of Variation						
	Variation	Variation %	DOF	Mean Square	F-Computed	F-Table
SSY	35.6400		45			
SSO	26.0969		1			
SSA (Overhead Var):	0.0019	0.0203	2	0.0010	15.592	6.940
SSB (Offered Load Var):	9.5376	99.9427	2	4.7688	76652.265	6.940
SSAB (Interaction):	0.0013	0.0135	4	0.0003	5.181	6.940
SSE (Error Var):	0.0022	0.0235	36	0.0001		
SST	9.5431		44	0.2169		

4.5.2 Effective Throughput Analysis. The effective throughput, as seen in Figure 4.10 above, is 98 % (16949 bps/17280 bps) of the maximum effective throughput at the 30 % offered load level. At the 60 % offered load level the throughput percentage is 99.8 % (34,491 bps/34,560 bps), and at the 90 % offered load level the throughput percentage is 99.9 % (51,798 bps/51,840 bps). As previously mentioned, the effective throughput will at times be less than, or even greater than, the ideal throughput due to the error in the exponential distribution of the IP packets not being delivered, on average, exactly one second apart.

The variation shown in Table 4-4 shows that 99.83 % of the variation is due to change in the offered load. The change in overhead effects 0.03 % of variation, and 0.13 % of the variation is due to error. The F-Computed valued of 13544.72 for the offered load is well above the F-Table value, verifying it has a significant impact on Link-16 network performance. The overhead and interaction variations have no effect on Link-16 performance as verified by smaller F-Computed values as compared to the F-Table values.

Table 4.4: Effective Throughput Allocation of Variation (ANOVA)

Table B.6: Effective Throughput Allocation of Variation						
	Variation	Variation %	DOF	Mean Square	F-Computed	F-Table
SSY	63345625271		45			
SSO	54302809601		1			
SSA (Overhead Var):	3032934	0.03	2	1516467	4.55	6.94
SSB (Offered Load Var):	9027475902	99.83	2	4513737951	13544.72	6.94
SSAB (Interaction):	309939	0.00	4	77485	0.23	6.94
SSE (Error Var):	11996896	0.13	36	333247		
SST	9042815670		44	205518538		

4.5.3 Raw Throughput Analysis. Figure 4.11 shows the raw throughput for the all three offered loads and all three overhead configurations. The raw throughput is actually greater than the number of data bits offered at the host terminal because of the overhead bits added to the data load before the JTIDS terminal transmits the JTIDS word.

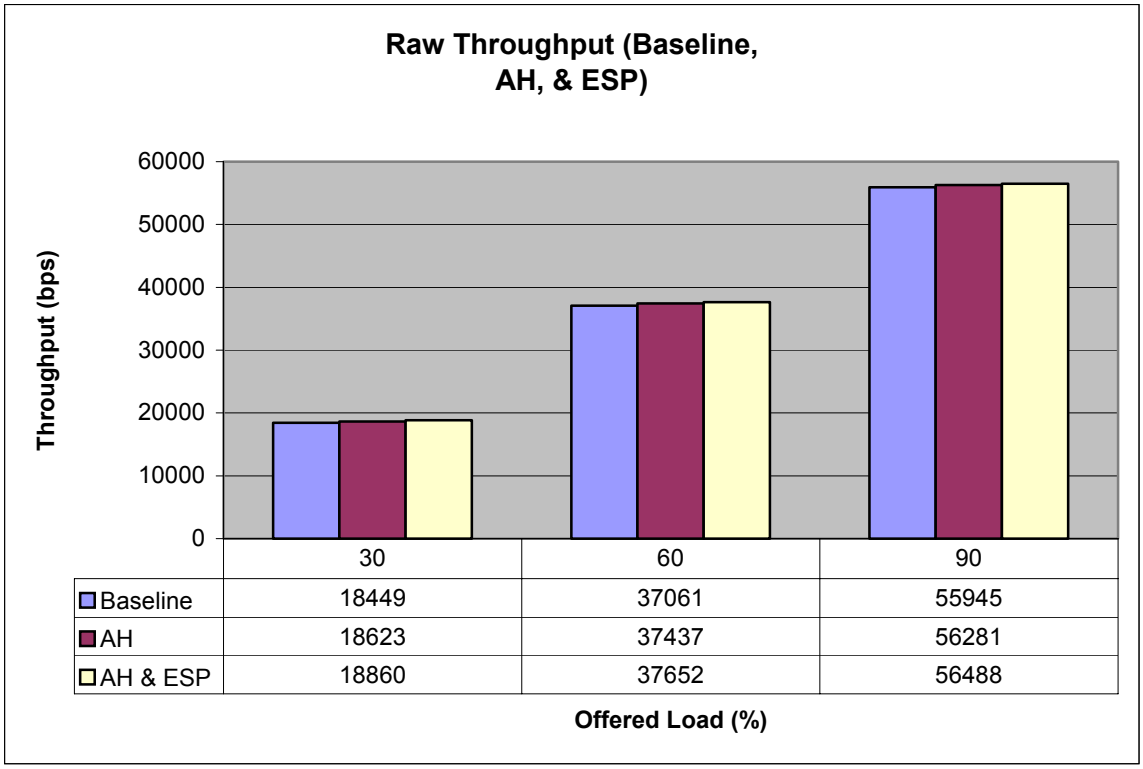


Figure 4.11: Baseline, Authentication Header, and Encapsulating Security Payload Raw Throughput

4.6 Confidence Interval(CI) Analysis

CI analysis can be used to determine if a measured value is significantly different from zero. The analysis is performed by checking the CI interval to see if it includes the value zero. If the CI includes zero, then the measured values are not significantly different than zero. Appendix A (Tables A.7 and A.8) and Appendix B (Tables B.7 and B.8) show the CI for the ETE delay and throughput offered load and overhead effects. The CI analysis shows that the results from the varied offered loads are significant, but the results from the varied security levels are not significant.

4.7 Summary

This chapter described the verification of the OPNET® Link-16 model and the resultant data from that model. Next, the implementation and results of the IP baseline model used for this research were explained. Then a performance summary of the overhead and workload variations was described. Finally an analysis of the results and data variations was explained.

V. Conclusions and Future Work

5.1 Overview

The JBI concept is within reach of becoming reality now that information technologies, in particular Internet technologies, are reaching a quality-of-service level that provides sufficient bandwidth for fast-paced war-fighting requirements. As various communication sub-systems are pieced together to form a network-centric JBI, Information Assurance becomes a chief concern. IPSec provides another security layer in the overall JBI security scheme. The research goal was to determine the feasibility of passing IP messages and IPSec messages over a Link-16 network. Sub-goals included performance analysis by comparing the effect that additional overhead from the IPSec security protocols has on a Link-16 network. In addition, various workloads were placed on the network to see how overall ETE delay and throughput are affected.

5.2 Conclusions

Since the IPSec security protocols add minimal overhead to the IP packet, and even though the Link-16 network is relatively slow (57.6 kbps) compared to most Internet pipelines, it has minimal effect on Link-16 ETE delay and throughput. Similar to varying the security levels, varying the offered load has a minimal effect on effective throughput. At the 60 % offered load level, effective throughput is 99.8 % of the maximum capacity. At the 90 % offered load level, effective throughput is 99.9 % of maximum capacity. Here we see, that an offered load through the 90 % level does not negatively impact the performance of the Link-16 network. However, the Link-16 data rates are still not adequate to for transferring large data files that typically traverse the Internet. Some research has been done, suggesting the Link-16 network can operate up to 1.0 Mbps. An effective throughput near the 1.0 Mbps level would be adequate to process large imagery files which typically range in size from 2 to 5 megabytes. At an average effective

throughput rate of 1.0 Mbps, a typical imagery file of 3 to 5 megabytes can take 3 to 5 seconds to download.

5.3 Contributions

This research demonstrates that although the IPSec security protocols contribute minimal to IP packet overhead, the main problem is that the Link-16 network is a legacy system that is far too slow to transfer large size data files. The AH and ESP security protocols can be added to the IPv6 packets with minimal effect on ETE delay and effective throughput. However, as offered load increases beyond 6.5 kbps, throughput begins to degrade rapidly. Therefore, the Link-16 network can transmit IP packets provided the average bytes per second rate doesn't exceed 6.5 kbps. The addition of IPSec protocols has minimal impact on the Link-16 network and does not degrade its performance.

5.4 Future Work

Since the USAF Tactical Datalink Roadmap has determined that Link-16 will be around for long term use, future work and research will always be useful to discover methods of improving the Link-16 data rate, such as software modifications or compression techniques. Further research can be conducted in this area by adjusting the Link-16 data rates to current maximum capacity of 238 kbps to determine effective throughput at that rate. Hardware modifications can also be made which effect the carrier modulation to increase the number of words in the packing structure.

It should be noted that the OPNET® Link-16 model used in this research was one of the early versions of several planned iterations, and some features such as relay and net stacking, are

not part of the current model set. Possible future research may include the use of multiple nets to simultaneously transmit data to increase throughput. It would be beneficial to validate the OPNET® Link-16 model used in this research work. However, real-time data is needed to validate results obtains from the Link-16 model.

Appendix A. End-to-End Delay Allocation of Variation (ANOVA) Worksheet

Table A.1: End-To-End Delay Data			
Offered Load (%)	**Baseline**	**AH**	**AH & ESP**
30	**17280 bits**	**17504 bits**	**17712 bits**
	0.254309	0.257448	0.261042
	0.252114	0.256790	0.258394
	0.258622	0.253870	0.254543
	0.248114	0.258590	0.263156
	0.264652	0.257132	0.256534
60	**34560 bits**	**34784 bits**	**34992 bits**
	0.648327	0.658174	0.662515
	0.647106	0.667412	0.657479
	0.663169	0.657781	0.672983
	0.659590	0.653576	0.657880
	0.640772	0.650811	0.663237
90	**51840 bits**	**52064 bits**	**52272 bits**
	1.339830	1.352140	1.392471
	1.366074	1.365532	1.383270
	1.347827	1.377320	1.388572
	1.371586	1.373314	1.390297
	1.342722	1.378324	1.383547

Table A.2: End-To-End Delay Mean			
Offered Load (%)	**Baseline**	**AH**	**AH & ESP**
30	0.256	0.257	0.259
60	0.652	0.658	0.663
90	1.354	1.369	1.388

Table A.3: End-To-End Delay Standard Deviations			
Offered Load (%)	**Baseline**	**AH**	**AH & ESP**
30	0.006343	0.001754	0.003442
60	0.009296	0.006303	0.006255
90	0.014321	0.010846	0.004096

Table A.4: End-To-End Delay Computation of Effects						
Offered Load (%)	**Baseline**	**AH**	**AH & ESP**	**Row Sum**	**Row Mean**	**Row Effect**
30	0.255562	0.256766	0.258734	0.771062	0.25702068	-0.504511482
60	0.651793	0.657551	0.662819	1.972162	0.65738741	-0.104144756
90	1.353608	1.369326	1.387631	4.110565	1.3701884	0.608656238
Column Sum	2.260963	2.283643	2.309184			
Column Mean	0.753654	0.761214	0.769728		0.761532	
Column Effect	-0.007878	-0.000318	0.008196			

Table A.5: End-To-End Delay Interaction Effects				
Offered Load (%)	**Baseline**	**AH**	**AH & ESP**	
30	0.006419	0.000063	-0.006483	0.000000
60	0.002283	0.000481	-0.002764	0.000000
90	-0.008703	-0.000545	0.009247	0.000000
	0.000000	0.000000	0.000000	

Table A.6: End-To-End Delay Allocation of Variation						
	Variation	**Variation %**	**DOF**	**Mean Square**	**F-Computed**	**F-Table**
SSY	35.6400		45			
SSO	26.0969		1			
SSA (Overhead Var):	0.0019	0.0203	2	0.0010	15.592	6.940
SSB (Offered Load Var):	9.5376	99.9427	2	4.7688	76652.265	6.940
SSAB (Interaction):	0.0013	0.0135	4	0.0003	5.181	6.940
SSE (Error Var):	0.0022	0.0235	36	0.0001		
SST	9.5431		44	0.2169		

Table A.7: Confidence Interval For Overhead Effects				
Parameter	**Mean Effect**	**Standard Dev.**	**C.I. (low end)**	**C.I. (high end)**
	0.7615	0.0012	0.7592	0.7638
Overhead				
Baseline	-0.0079	0.0017	-0.0111	-0.0046
AH	-0.0003	0.0017	-0.0036	0.0029
AH & ESP	0.0082	0.0017	0.0049	0.0115

Table A.8: Confidence Interval For Offered Load Effects				
Parameter	**Mean Effect**	**Standard Dev.**	**C.I. (low end)**	**C.I. (high end)**
	0.7615	0.0012	0.7596	0.7638
Offered Load				
30%	-0.5045	0.0017	-0.5078	-0.5013
60%	-0.1041	0.0017	-0.1074	-0.1009
90%	0.6087	0.0017	0.6054	0.6119

Appendix B. Effective Throughput Allocation of Variation (ANOVA) Worksheet

Table B.1: Effective Throughput Data			
Offered Load (%)	**Baseline**	**AH**	**AH & ESP**
30	**17280 bits**	**17504 bits**	**17712 bits**
	17073	17794	17132
	16829	16970	17968
	16540	16915	17395
	17215	18163	17927
	17088	17018	17487
60	**34560 bits**	**34784 bits**	**34992 bits**
	34277	34495	35395
	35544	35307	34635
	34101	35126	34874
	33664	35271	36167
	34868	34333	35738
90	**51840 bits**	**52064 bits**	**52272 bits**
	51573	52841	52334
	52803	51376	51756
	51785	52382	52027
	51947	51284	53258
	50883	52023	51628

Table B.2: Effective Throughput Mean			
Offered Load (%)	**Baseline**	**AH**	**AH & ESP**
30	16949	17372	17582
60	34491	34906	35362
90	51798	51981	52201

Table B.3: Effective Throughput Standard Deviations			
Offered Load (%)	**Baseline**	**AH**	**AH & ESP**
30	267.98	569.93	358.57
60	730.22	458.20	623.93
90	692.94	662.19	650.30

Table B.4: Effective Throughput Computation of Effects						
Offered Load (%)	**Baseline**	**AH**	**AH & ESP**	**Row Sum**	**Row Mean**	**Row Effect**
30	16949.00	17372.05	17581.89	51902.94	17300.98	-17437.03
60	34490.76	34906.35	35361.80	104758.91	34919.64	181.63
90	51798.42	51981.20	52200.60	155980.22	51993.41	17255.40
Column Sum	103238.17	104259.60	105144.29			
Column Mean	34412.72	34753.20	35048.10		34738.01	
Column Effect	-325.28	15.19	310.09			

Table B.5: Effective Throughput Interaction Effects				
Offered Load (%)	**Baseline**	**AH**	**AH & ESP**	
30	-26.70	55.88	-29.18	0.00
60	-103.60	-28.48	132.07	0.00
90	130.29	-27.40	-102.90	0.00
	0.00	0.00	0.00	

Table B.6: Effective Throughput Allocation of Variation						
	Variation	**Variation %**	**DOF**	**Mean Square**	**F-Computed**	**F-Table**
SSY	63345625271		45			
SSO	54302809601		1			
SSA (Overhead Var):	3032934	0.03	2	1516467	4.55	6.94
SSB (Offered Load Var):	9027475902	99.83	2	4513737951	13544.72	6.94
SSAB (Interaction):	309939	0.00	4	77485	0.23	6.94
SSE (Error Var):	11996896	0.13	36	333247		
SST	9042815670		44	205518538		

Table B.7: Confidence Interval For Overhead Effects				
Parameter	Mean Effect	Standard Dev.	C.I. (low end)	C.I. (high end)
	34738.01	86.06	34569.34	34906.67
Overhead				
Baseline	-325.28	121.70	-563.82	-86.75
AH	15.19	121.70	-223.34	253.73
AH & ESP	310.09	121.70	71.56	548.62

Table B.8: Confidence Interval For Offered Load Effects				
Parameter	Mean Effect	Standard Dev.	C.I. (low end)	C.I. (high end)
	34738.01	86.06	34596.45	34906.67
Offered Load				
30%	-17437.03	121.70	-17675.56	-17198.50
60%	181.63	121.70	-56.90	420.16
90%	17255.40	121.70	17016.87	17493.93

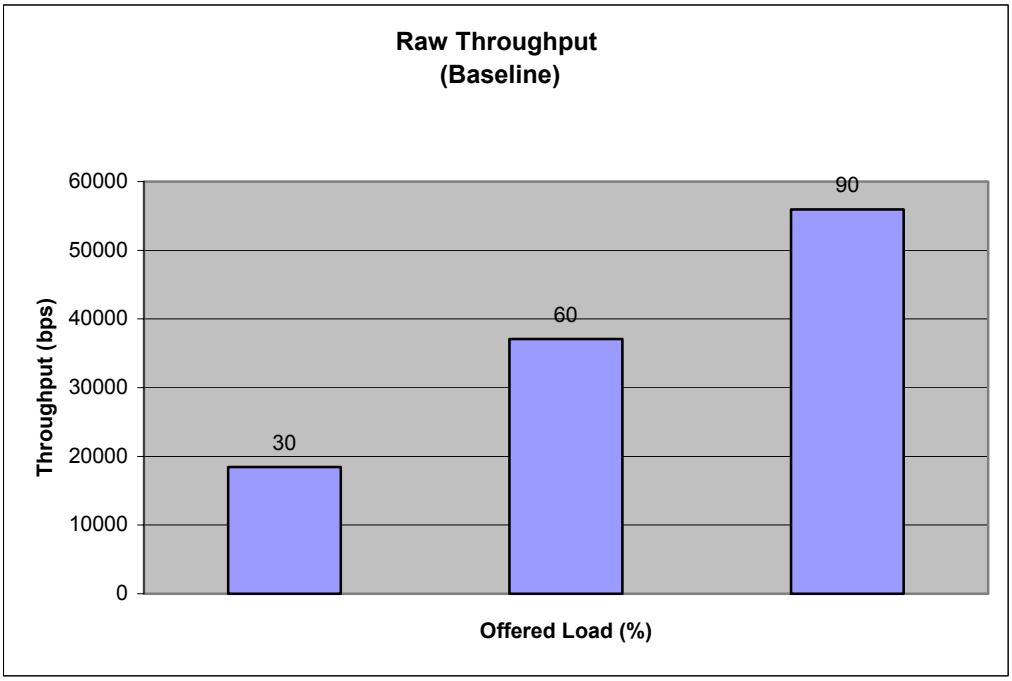
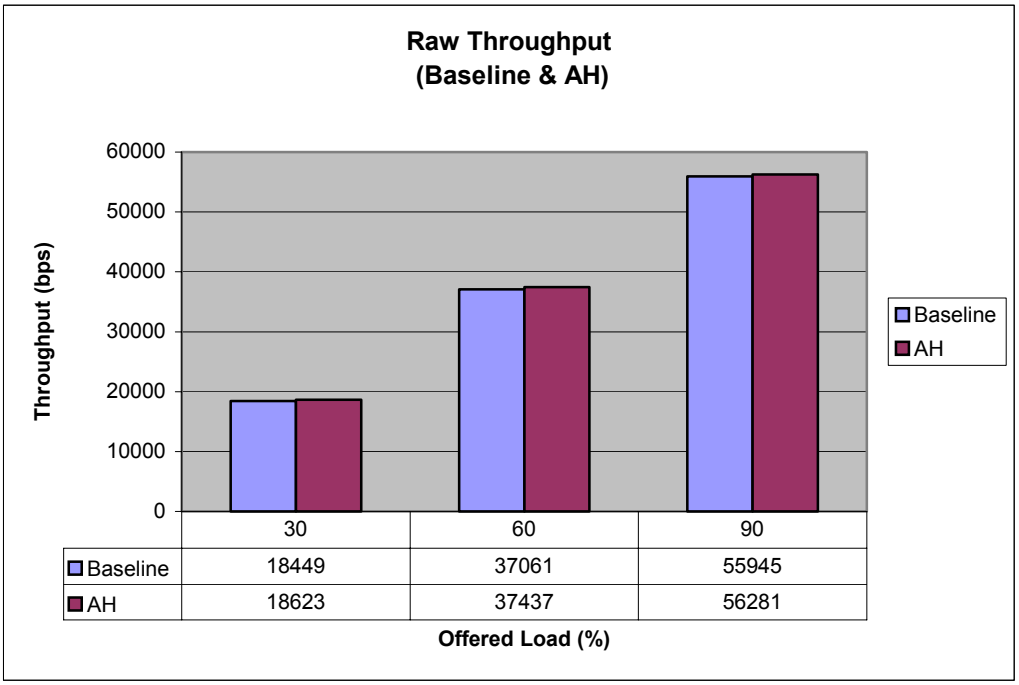


Figure C.1: Raw Throughput (Baseline)



	30	60	90
Baseline	18449	37061	55945
AH	18623	37437	56281

Figure C.2: Raw Throughput (Baseline and AH)

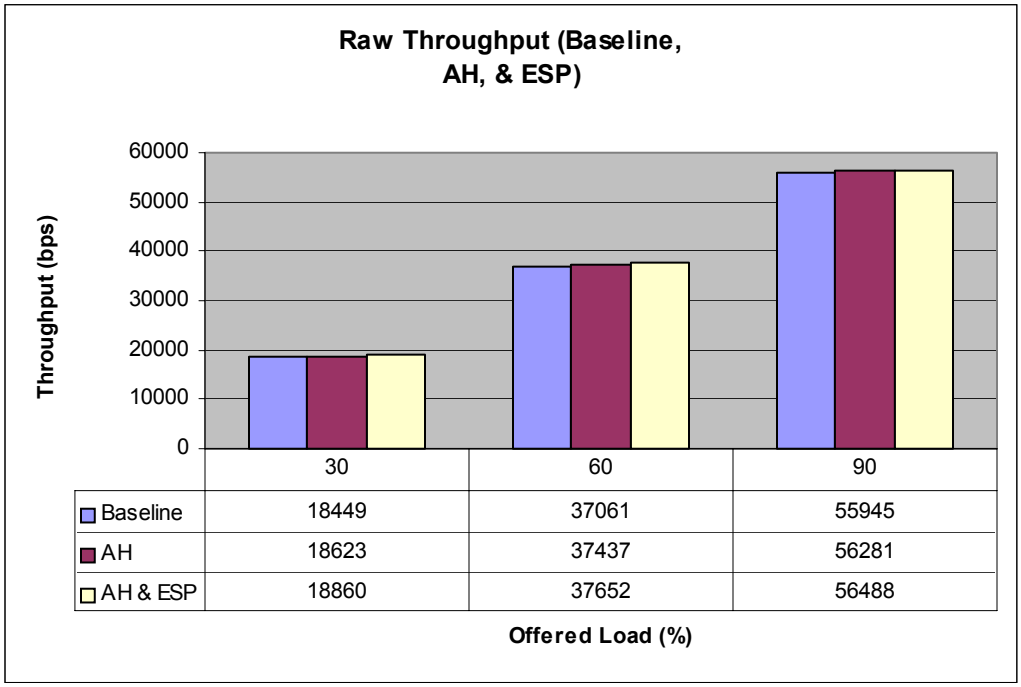


Figure C.3: Raw Throughput (Baseline, AH, and ESP)

Table D.1: Baseline		
30% Baseline		
95% C.I., z =	1.96	
Std. Dev:	0.006343	
Sample Mean:	0.255562	1.243184
Accuracy, r (%) =	5	1.277811
n =	0.9465	0.972901
60% Baseline		
95% C.I., z =	1.96	
Std. Dev:	0.022164	
Sample Mean:	0.469752	4.344159
Accuracy, r (%) =	5	2.348761
n =	3.4208	1.849553
90% Baseline		
95% C.I., z =	1.96	
Std. Dev:	0.039549	
Sample Mean:	1.060222	7.751524
Accuracy, r (%) =	5	5.30111
n =	2.1382	1.462245

Table D.2: Authentication Header (AH)		
30% AH		
95% C.I., z =	1.96	
Std. Dev:	0.006455	
Sample Mean:	0.273497	1.265097
Accuracy, r (%) =	5	1.367486
n =	0.8559	0.925126
60% AH		
95% C.I., z =	1.96	
Std. Dev:	0.009691	
Sample Mean:	0.469752	1.899386
Accuracy, r (%) =	5	2.348761
n =	0.6540	0.808676
90% AH		
95% C.I., z =	1.96	
Std. Dev:	0.012579	
Sample Mean:	1.060222	2.465449
Accuracy, r (%) =	5	5.30111
n =	0.2163	0.465082

Table D.3: AH & ESP		
30% AH & ESP		
95% C.I., z =	1.96	
Std. Dev:	0.013140	
Sample Mean:	0.255562	2.575354
Accuracy, r (%) =	5	1.277811
n =	4.0620	2.015442
60% AH & ESP		
95% C.I., z =	1.96	
Std. Dev:	0.013571	
Sample Mean:	0.514804	2.659843
Accuracy, r (%) =	5	2.574022
n =	1.0678	1.033341
90% AH & ESP		
95% C.I., z =	1.96	
Std. Dev:	0.007291	
Sample Mean:	1.091631	1.429121
Accuracy, r (%) =	5	5.458157
n =	0.0686	0.261832

```
range = 1200;

rand('seed', 1);
L0 = zeros(1,range);
L0(1) = randexpo(1,1);
for i = 2:range
   L0(i) = L0(i-1)+randexpo(1,1);
end

L1 = zeros(1,range);
L1(1) = randexpo(1,1);
for i = 2:range
   L1(i) = L1(i-1)+randexpo(1,1);
end

L2 = zeros(1,range);
L2(1) = randexpo(1,1);
for i = 2:range
   L2(i) = L2(i-1)+randexpo(1,1);
end

L3 = zeros(1,range);
L3(1) = randexpo(1,1);
for i = 2:range
   L3(i) = L3(i-1)+randexpo(1,1);
end

L4 = zeros(1,range);
L4(1) = randexpo(1,1);
for i = 2:range
   L4(i) = L4(i-1)+randexpo(1,1);
end

L5 = zeros(1,range);
L5(1) = randexpo(1,1);
for i = 2:range
   L5(i) = L5(i-1)+randexpo(1,1);
end

L6 = zeros(1,range);
L6(1) = randexpo(1,1);
for i = 2:range
```

```
L6(i) = L6(i-1)+randexpo(1,1);
end

LO=L0'
L1=L1'
L2=L2'
L3=L3'
L4=L4'
L5=L5'
L6=L6'
%rand('seed', 109);

  save c0.dat L0 -ASCII
  save c1.dat L1 -ASCII
  save c2.dat L2 -ASCII
  save c3.dat L3 -ASCII
  save c4.dat L4 -ASCII
  save c5.dat L5 -ASCII
  save c6.dat L6 -ASCII

%rand('seed', 1109);
```

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Vita

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14. ABSTRACT The purpose of Link-16 is to exchange real-time tactical data among units of the United States and allied forces. Primary Link-16 functions include exchange of friendly unit position and status data, the dissemination of tactical surveillance track data, and the control/management of air, surface, and subsurface engagements. Because Link-16 will play an integral role in the network-centric Joint Battlespace Infosphere (JBI), the performance of Internet Protocol version six (IPv6) and IP Security (IPSec) over Link-16 needs to be determined. Using OPNET® modeling software to simulate a Link-16 network, the investigation of this research revealed that the overhead from IPv6 and IPSec does not significantly affect end-to-end delay and effective throughput of the Link-16 network. As long as the encryption and authentication protocols are preprocessed, these protocols add minimal amounts of latency overhead to the Link-16 network. However, as the offered load is extended beyond the 90% level, the overhead from the IPSec extensions begins to have more of a negative effect on the End-to-End delay and throughput. Therefore, as the offered load increases beyond the 90% level, it begins to have a significant impact on the performance of the Link-16 network.

15. SUBJECT TERMS

Link-16, TADIL-J, Common Data Link, Network Security, Information Assurance, Internet Protocol

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